

# Single-Cell Dissection of Healthy vs Malignant Bone Marrow Microenvironments

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02 Data & Workflow

03 Quality Control & Differential Expression

04 Dimensionality reduction & Clustering

05 Cell Type Annotation

06 Cell-Cell Communication

07 Pathway Spotlights

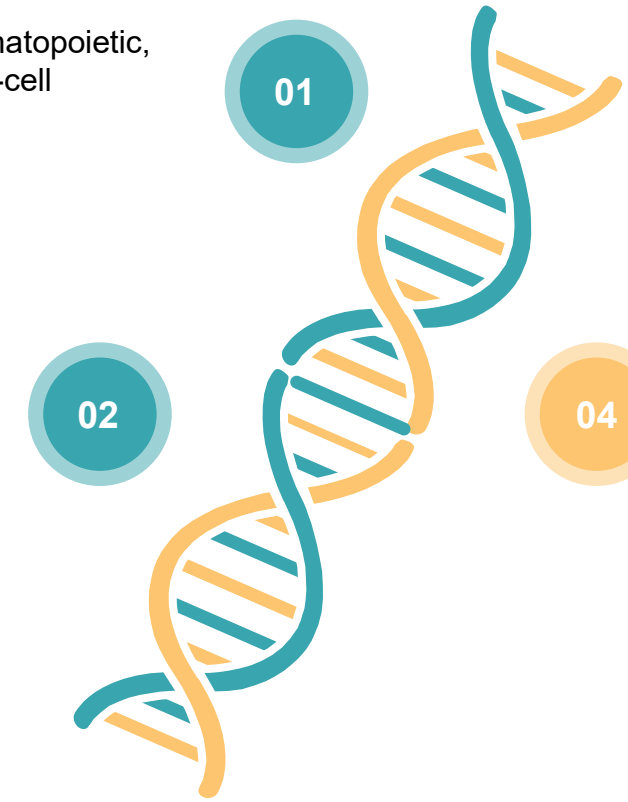
08 Biological Insights

# Background & Rationale

# Background & Rationale

**Bone-marrow microenvironment**

Primary site of hematopoiesis; composed of hematopoietic, immune, and stromal cells that coordinate blood-cell production.




**Limitations of bulk profiling**

Conventional RNA-seq averages signals across cell types, masking rare or lineage-specific changes.

**Impact of malignancy**

Myeloma cells remodel this niche, rewiring cell-cell signaling to promote tumor survival and immune evasion.

**Power of ScRNA-seq**

Resolves cellular heterogeneity and uncovers transcriptional states at single-cell resolution.



# Why this study?

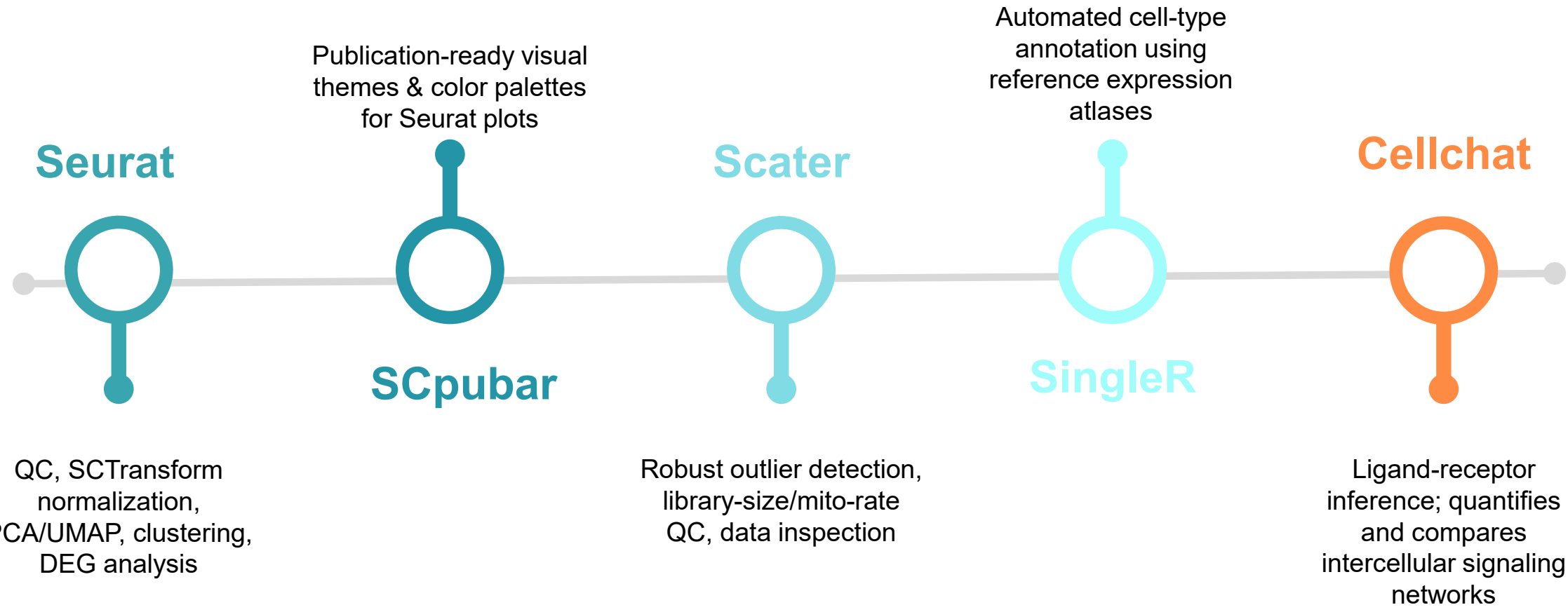
Side-by-side comparison of healthy vs. malignant bone marrow will pinpoint disrupted pathways, highlight candidate therapeutic targets (e.g., immunosuppressive signals), and deepen our understanding of tumor-niche crosstalk.

## Objectives :

- Compare transcriptomes of healthy vs malignant bone-marrow cells.
- Identify differentially expressed genes (DEGs) & dysregulated pathways.
- Identify and annotate different cell types (SingleR)
- Map intercellular communication networks (CellChat).


Data & Workflow

# Key Packages and roles



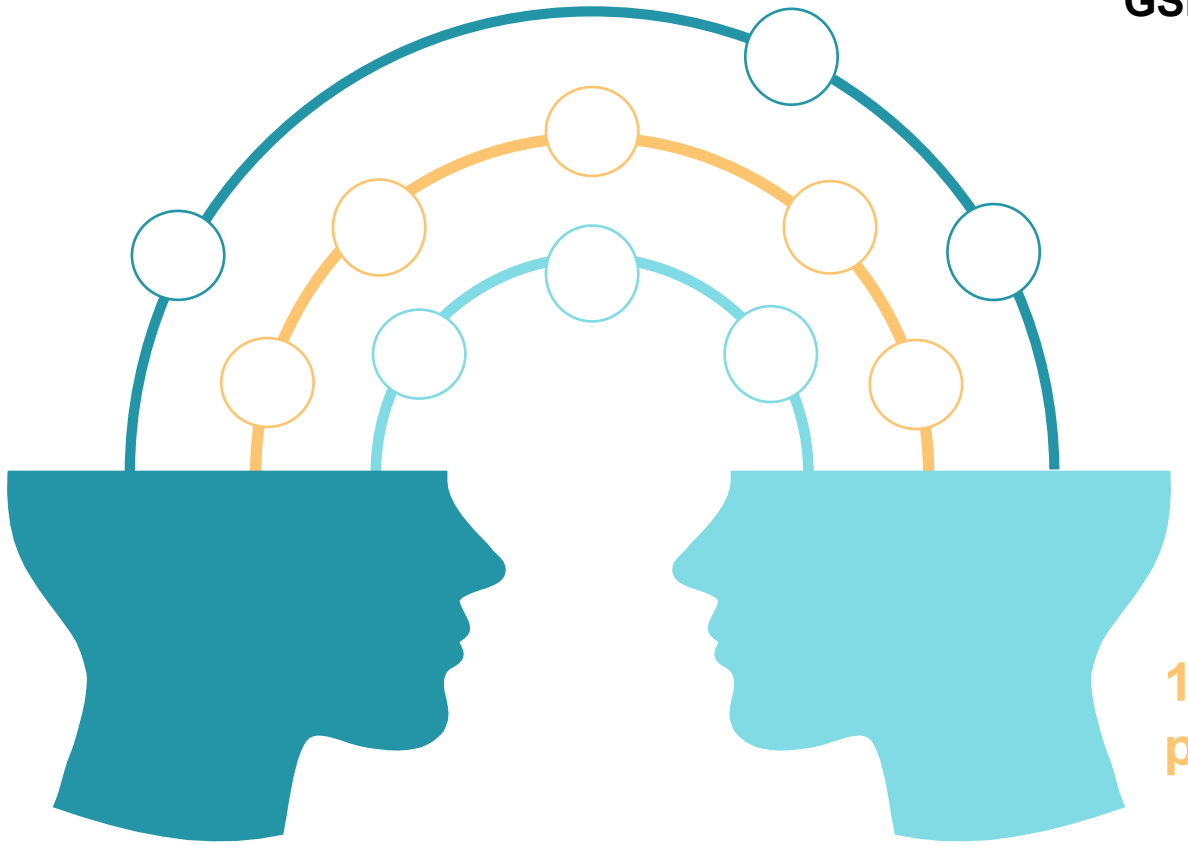
# Data set & Samples

**GEO accession number**

**GSE176131**

**Healthy sample**

**GSM5356722**



**Malignant sample**

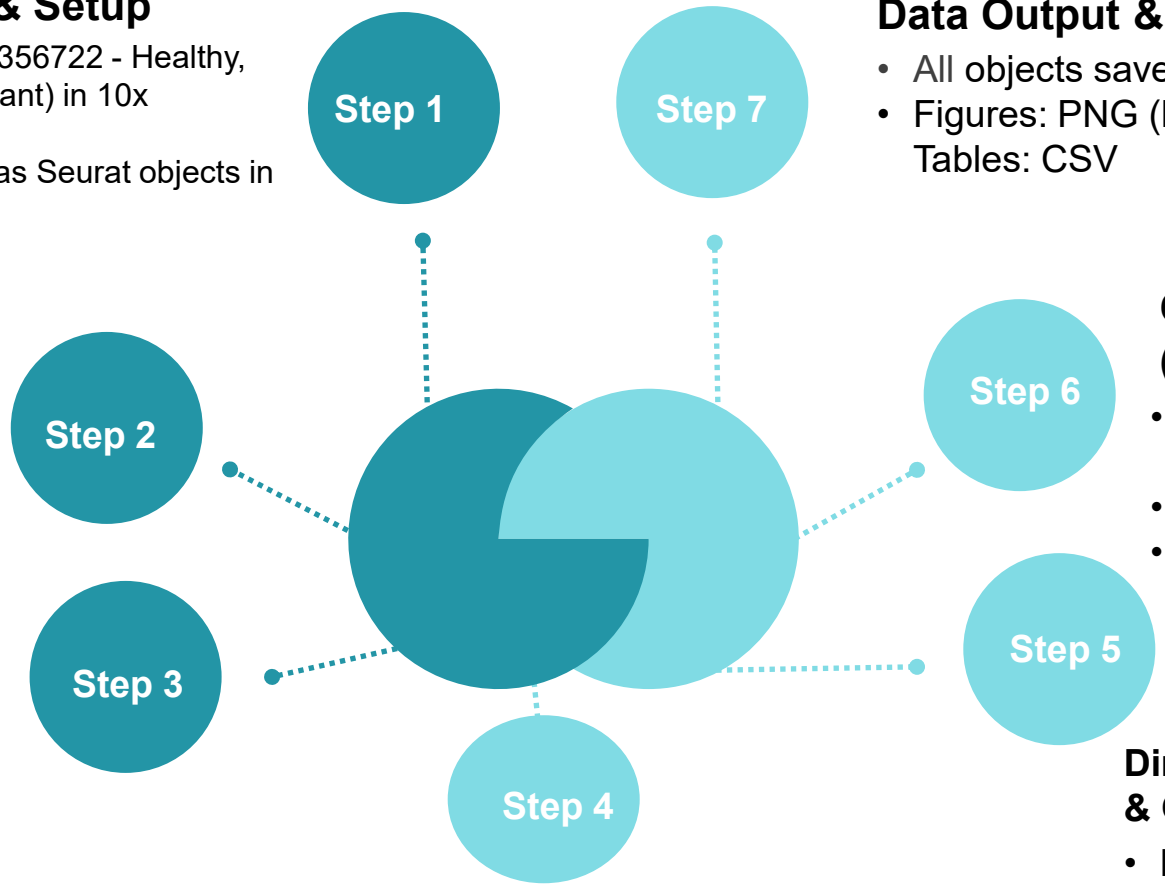
**GSM5356726**

**10x Genomics 3′ GEX; processed in Seurat 5.**

# Analysis workflow

## Data Acquisition & Setup

- GEO Datasets (GSM5356722 - Healthy, GSM5356726 - Malignant) in 10x Genomics format.
- Imported and merged as Seurat objects in R.




## Data Output & Management

- All objects saved as RDS
- Figures: PNG (high resolution), Tables: CSV

## Quality Control (QC)

- Cell-level: library size, feature count, mitochondrial %.
- Gene-level: remove genes expressed in <3 cells.
- Final: ~18,546 high-quality cells retained.

## Cell-Cell Communication (CellChat)

- Constructed interaction networks for Healthy vs Malignant
- Compared pathway activity & centrality
- Visualizations: circle plots, bubble plots, heatmaps

## Pre-processing & Normalization

- Log-normalization, variable feature selection, and data scaling
- SCTransform: variance-stabilizing transformation & cell-cycle regression

## Dimensionality Reduction, Clustering & Cell Type Annotation

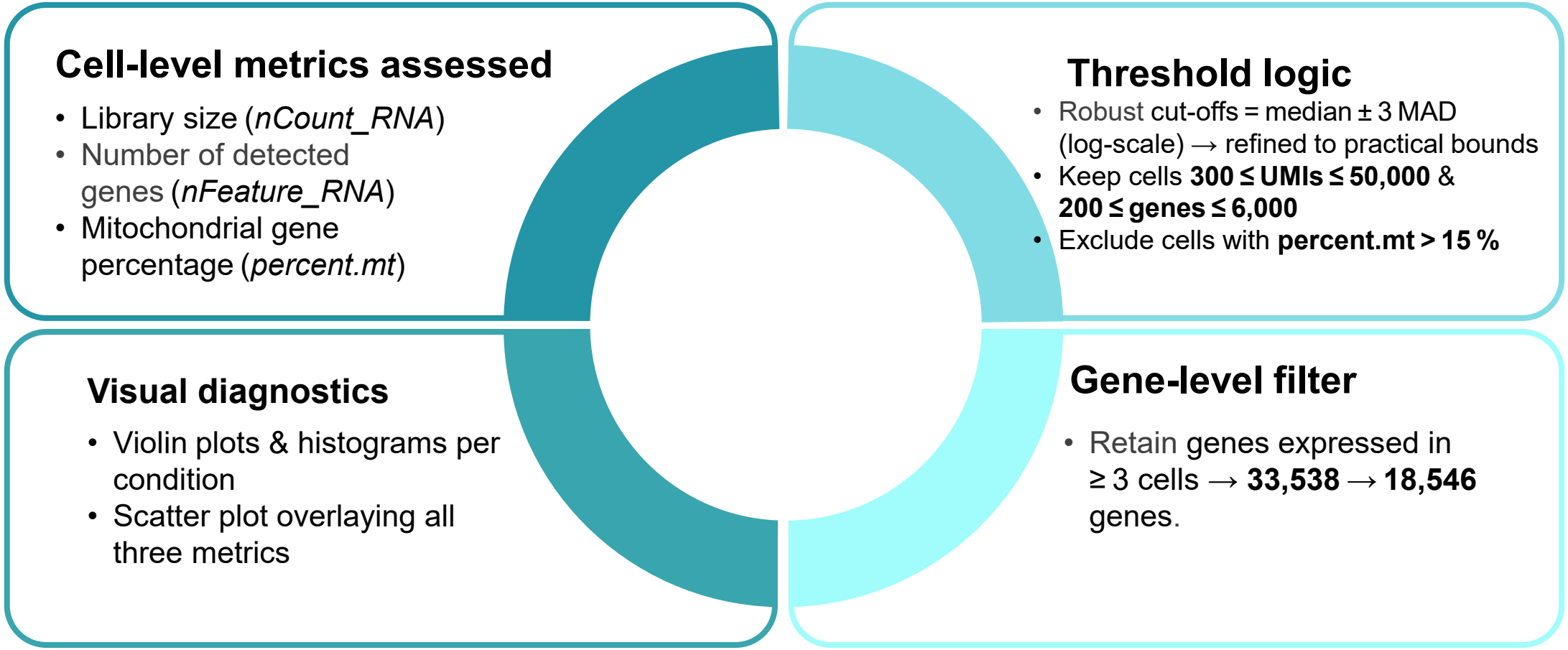
- PCA → Elbow plot selection → UMAP visualization
- Graph-based Louvain clustering (resolution 0.4)
- Annotated clusters using **SingleR**

## Differential Expression Analysis (DEA)

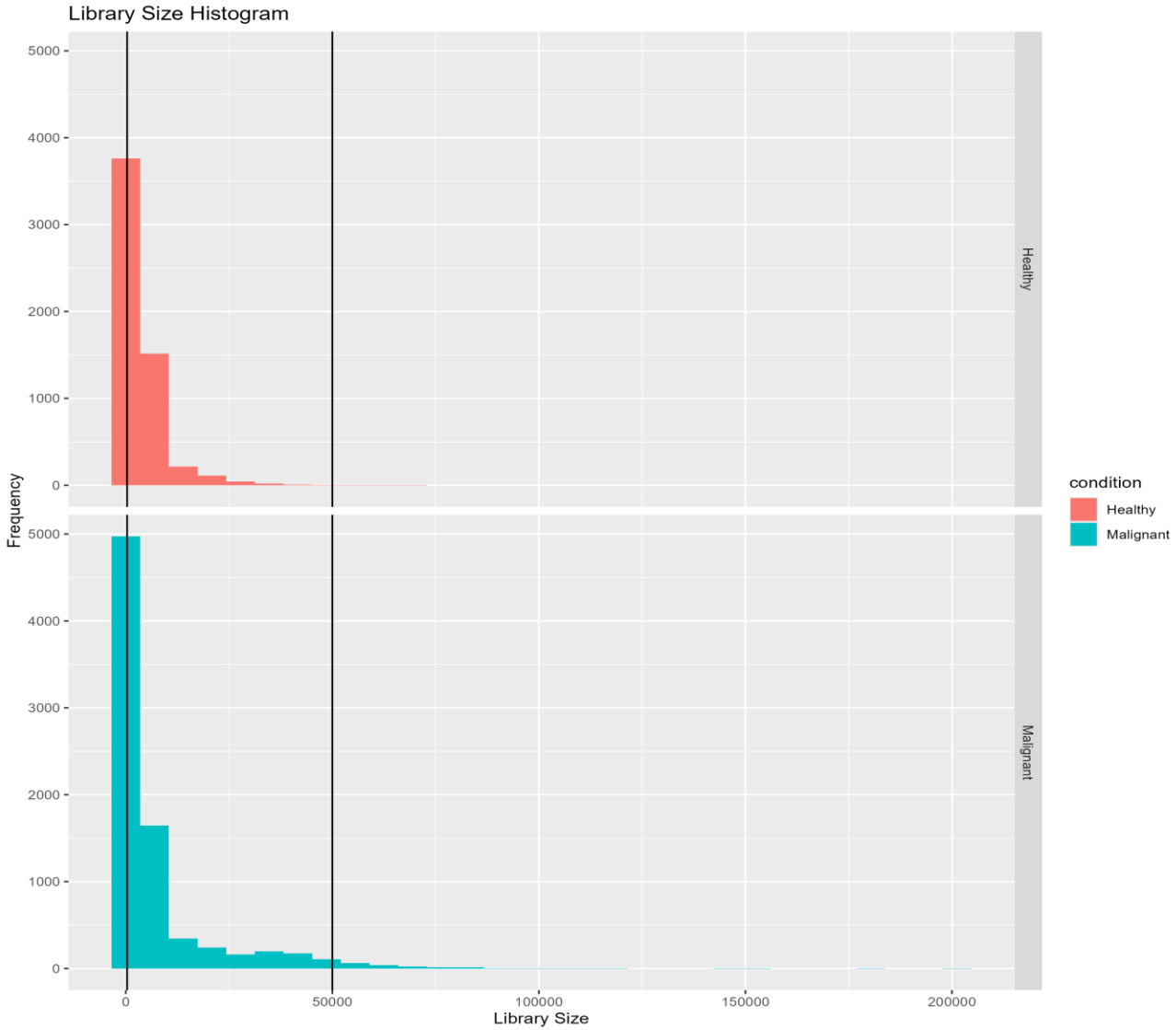
- Seurat Wilcoxon test: $\log_2FC > 0.25$, FDR < 0.05
- Volcano & heatmaps to visualize DEGs


Quality Control

# Quality-Control Strategy



# Library Size QC



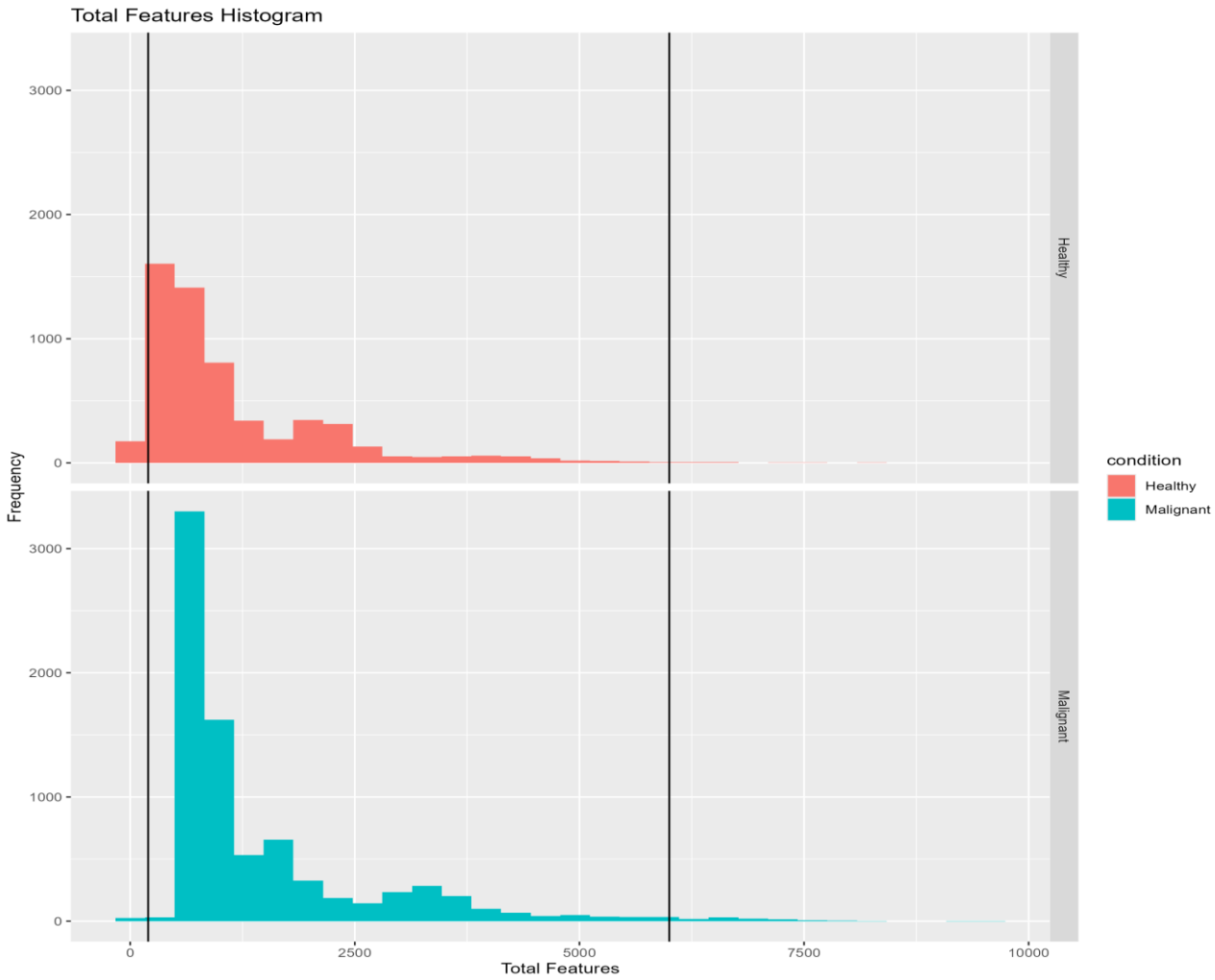

## Shape of the curves

**Healthy**: tight; 1k - 10k UMIs; very few > 20k → compact, regulated transcriptomes.
**Malignant:** right-skewed; long tail beyond 200 k UMIs → pronounced transcriptional heterogeneity.

## QC cut-offs

**< 300 UMIs** → exclude empty / low-quality droplets.
**50,000 UMIs** → exclude probable doublets / multiplets.

# Total Features QC




## Shape of the curves

•**Healthy:** sharp decline after **2,500 genes/cell** → compact, regulated transcriptomes.
•**Malignant:** extended right tail; many cells exceed **5,000 genes/cell** → higher complexity.

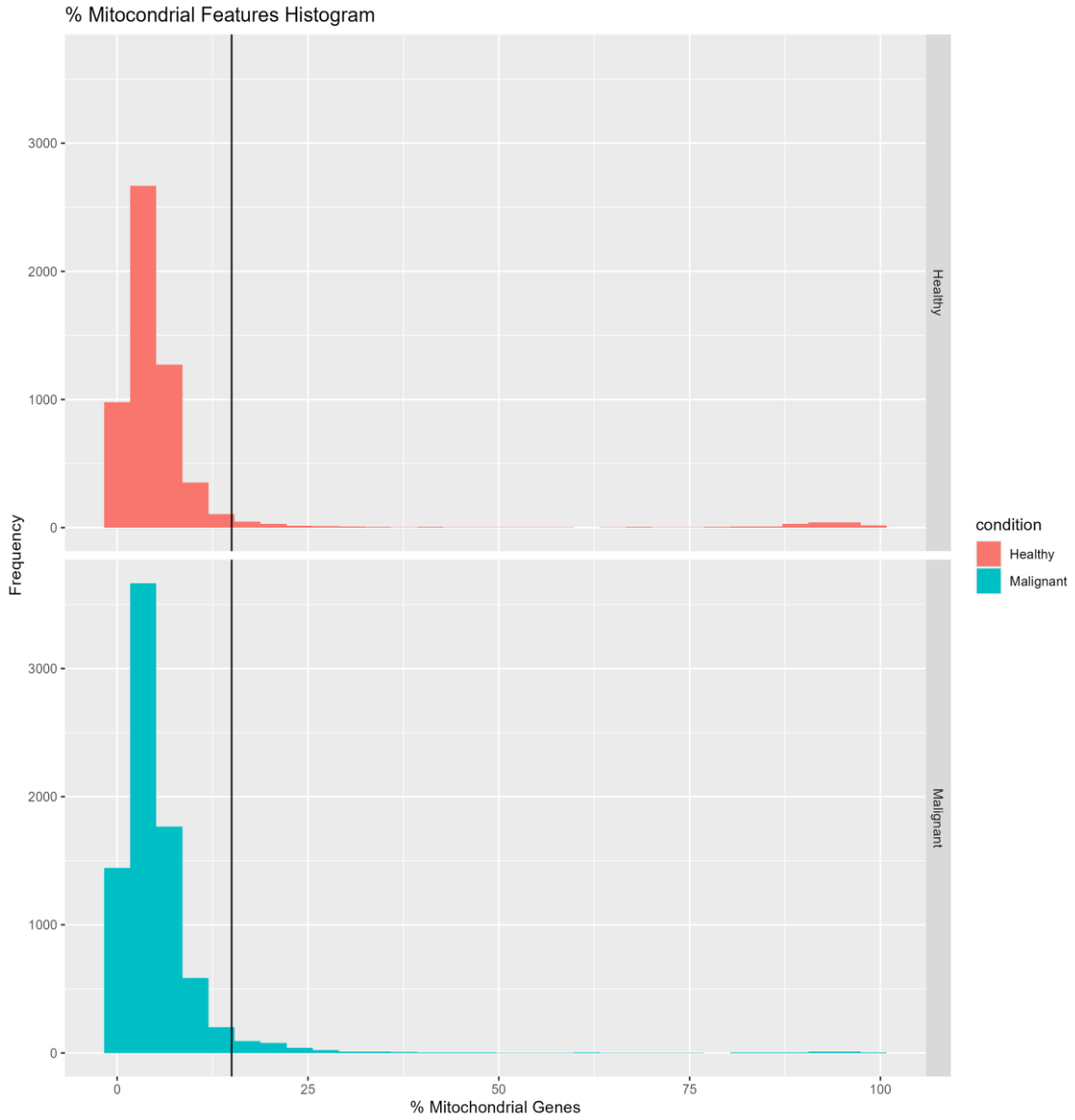
## QC cut-offs

•**< 200 genes** → remove low-quality / dying cells.
•**> 6,000 genes** → remove probable doublets or multiplets.

## Biological take-away

Malignant marrow exhibits greater transcriptomic complexity consistent with diverse tumor subclones and activated immune/stromal cells.

# Mitochondrial Content QC




## Shape of the curves

- Most cells < 15 % in both samples.
- Malignant sample shows broader tail; more cells with 15-30 % indicating metabolic stress.

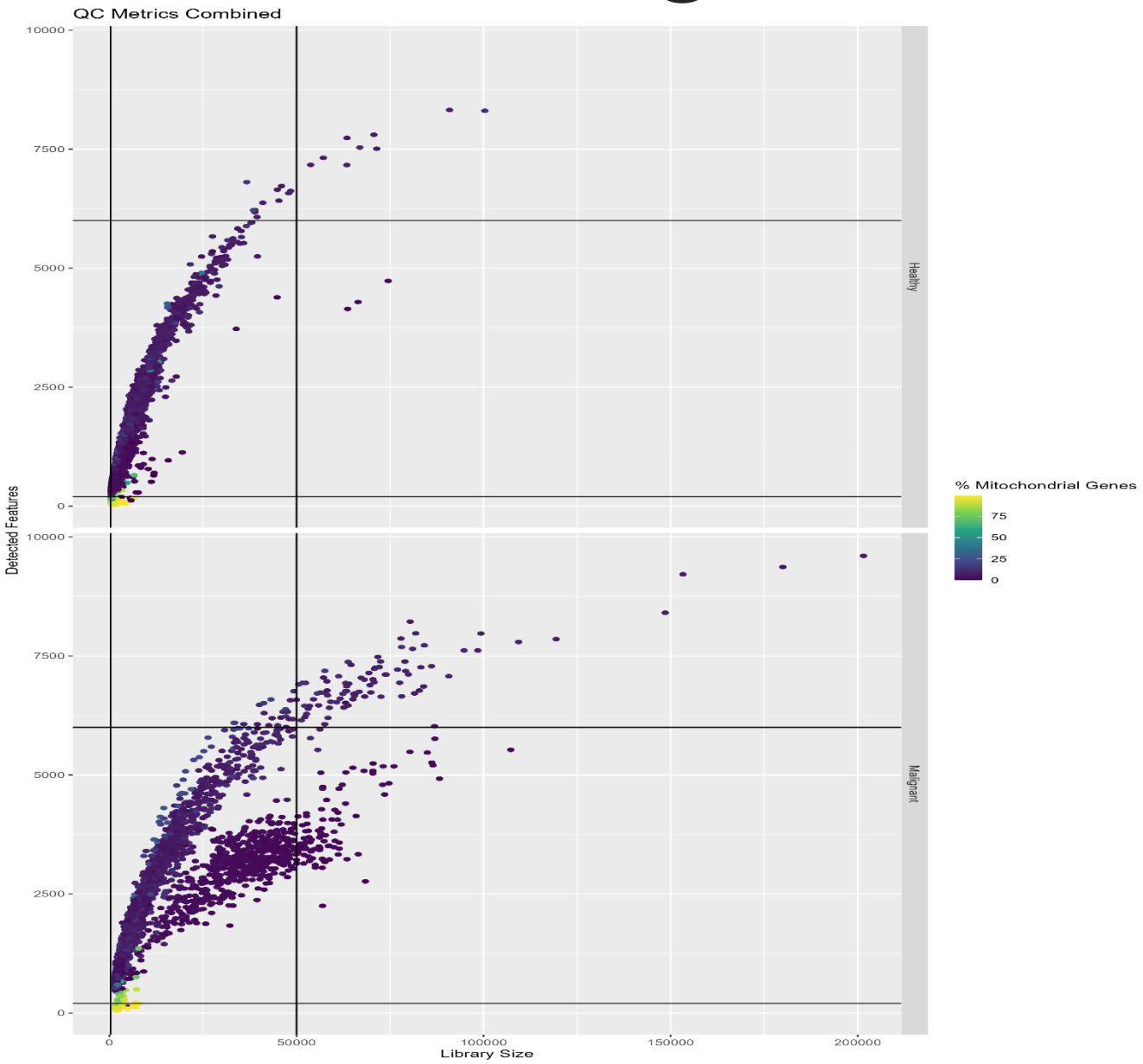
## QC rule

- Exclude cells **percent.mt > 15%**.

## Interpretation

- Elevated mitochondrial reads in malignant cells may reflect apoptosis, oxidative stress, or altered metabolism.

# Integrated QC Scatter

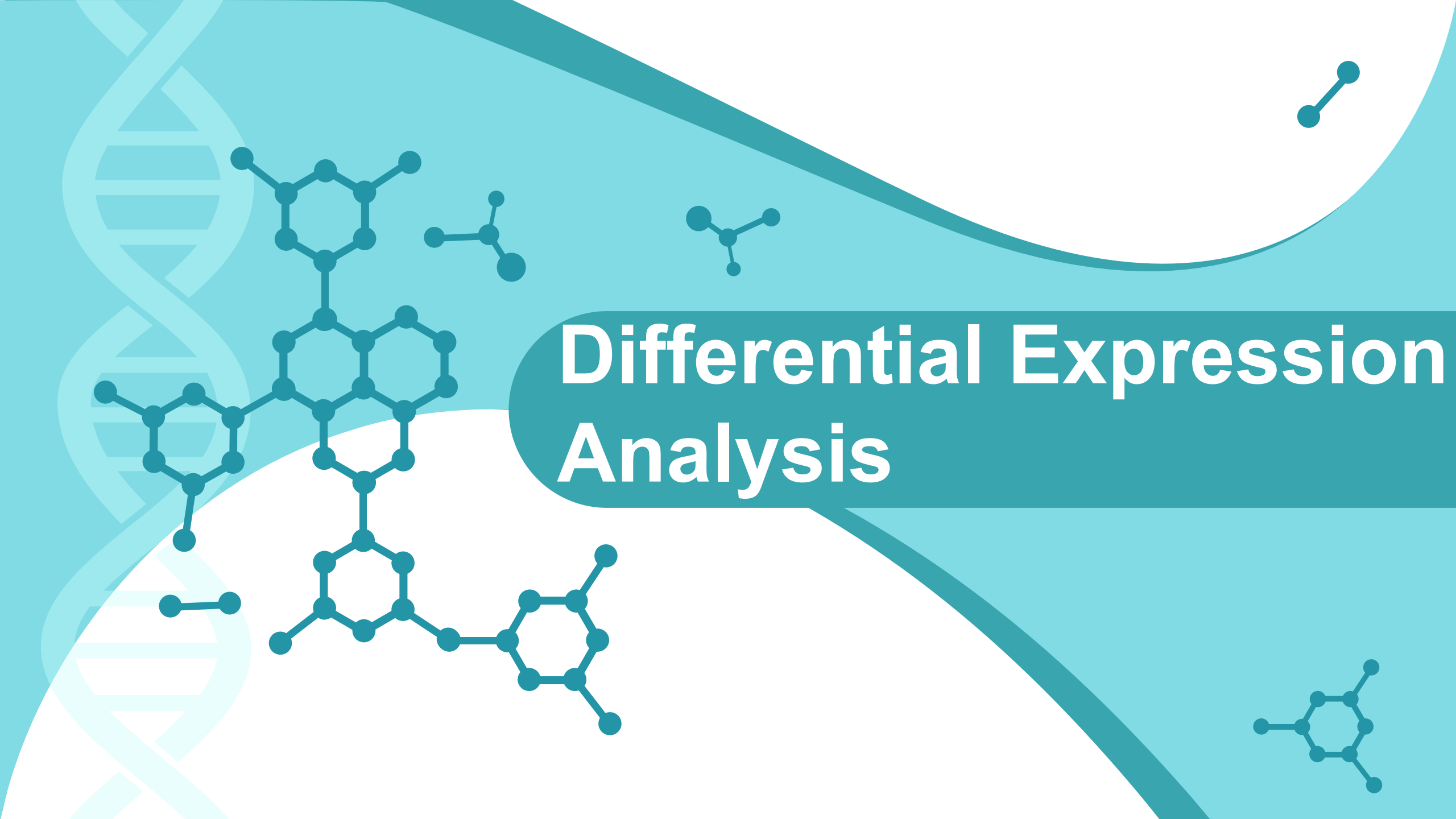



## Healthy panel

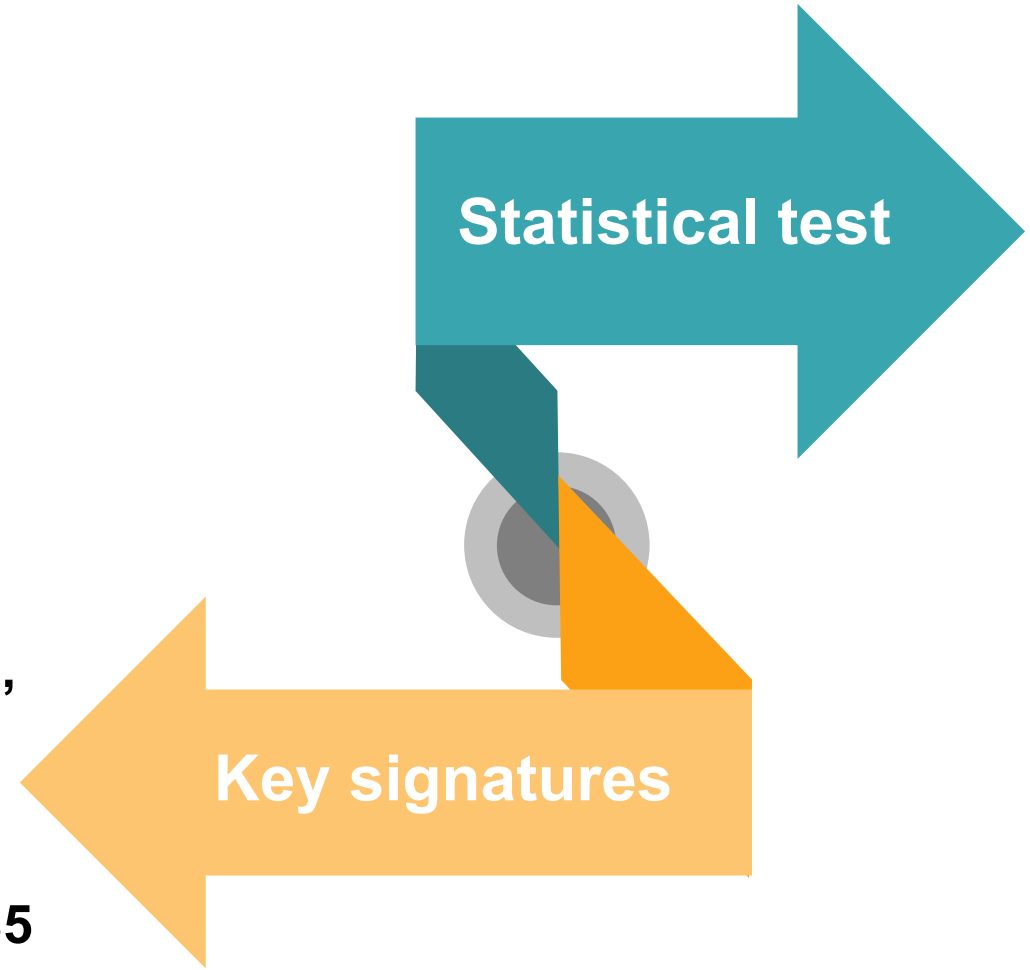
- Tight arc below QC limits → uniform library & feature counts
- Mostly low mito% (dark points); few yellow outliers
- Rare cells above 6k genes or 50k UMIs → likely doublets / multiplets

## Malignant panel

- Broader cloud; sub-set with very high UMIs/genes → clonal expansion
- Higher average mito%; several cells > 25 % (yellow) → stress/apoptosis
- Confirms greater transcriptional & metabolic heterogeneity in malignancy


Differential Expression
Analysis

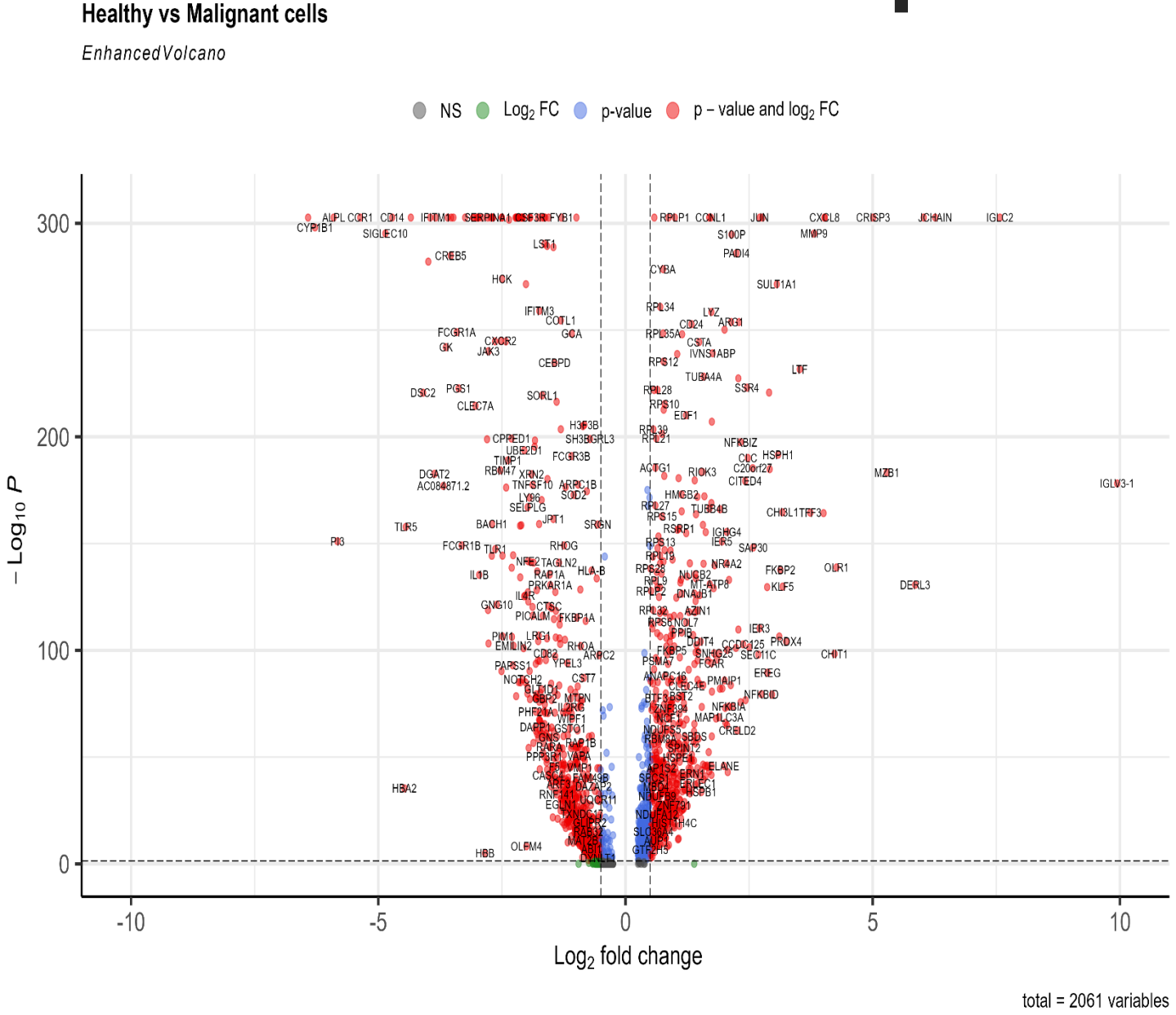
# Differential Expression Overview




- Wilcoxon rank-sum ($log_2FC \geq 0.25$ , $FDR < 0.05$)
- 2,061 significant DEGs
- ↑ Malignant: 1,017
  ↓ Malignant: 1,044

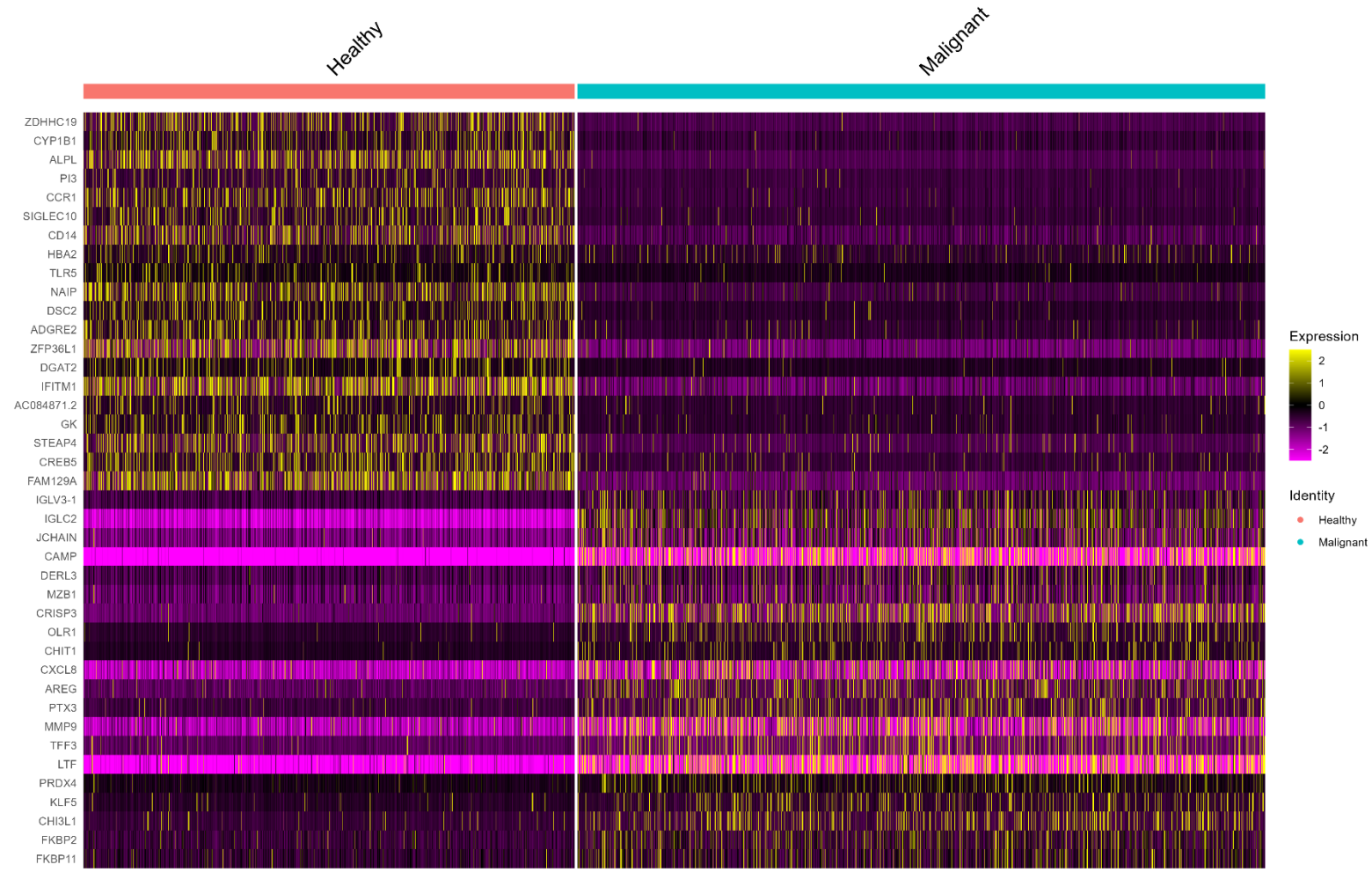
- **Up-regulated**
  IGLV3-1, IGLC2, JCHAIN, CAMP, DERL3, MZB1
- **Down-regulated**
  CD14, CYP1B1, ALPL, CCR1, SIGLEC10, CREB5

# Differential Expression Overview

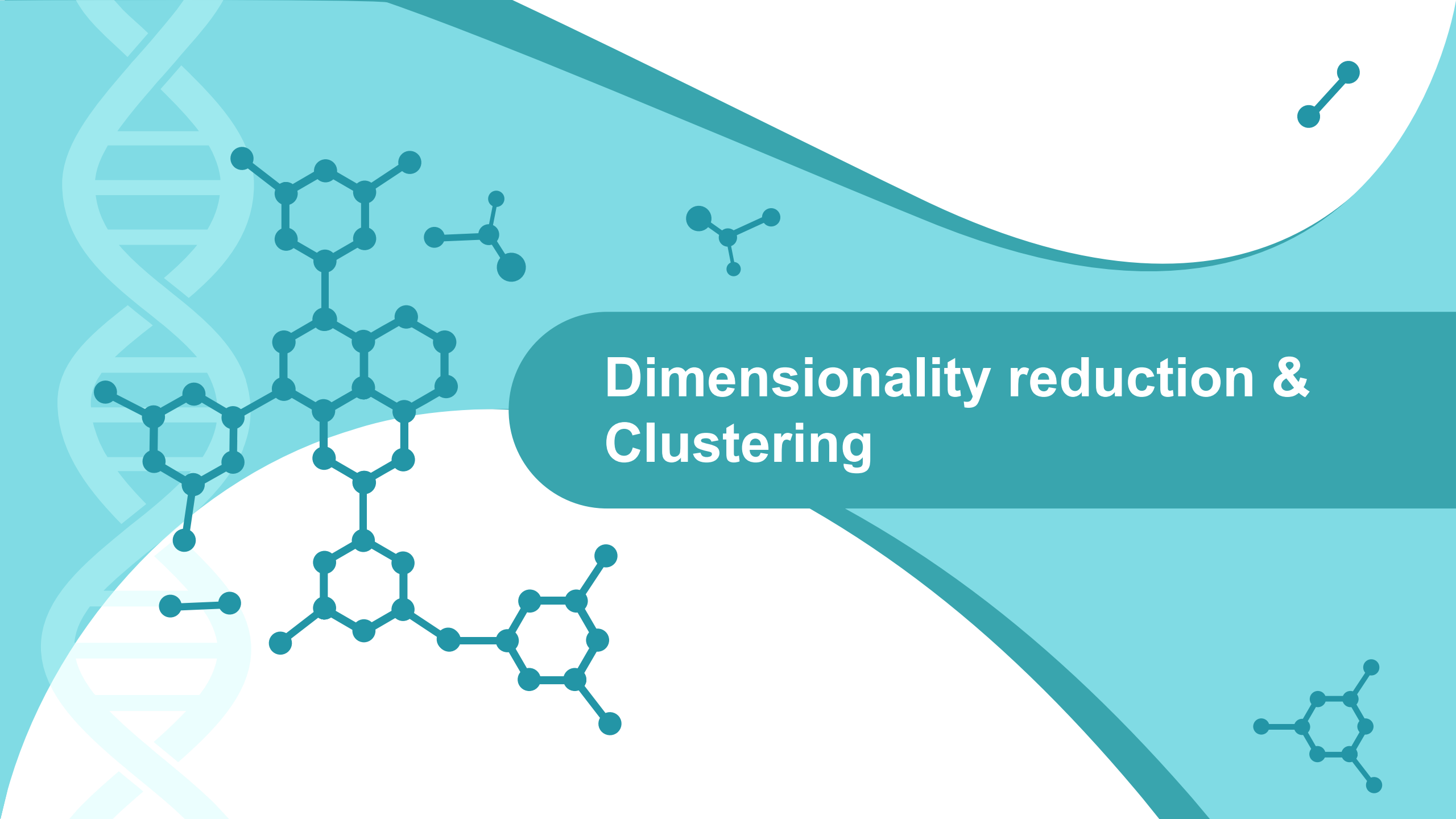



- **Balanced up/down**
- **plasma-cell markers dominate right flank**
- **Myeloid markers anchor left.**

# Differential Expression Overview

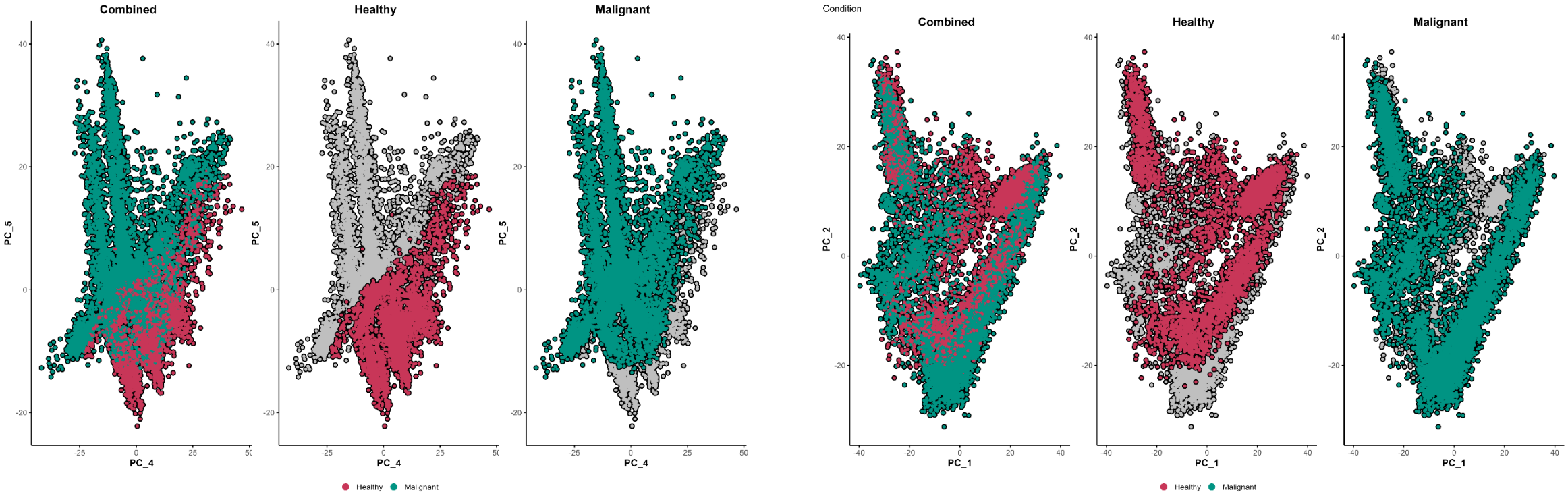



- **Heatmap (top 20 ↑ genes)** → clear Healthy vs Malignant separation; secretory & ER-stress genes enriched

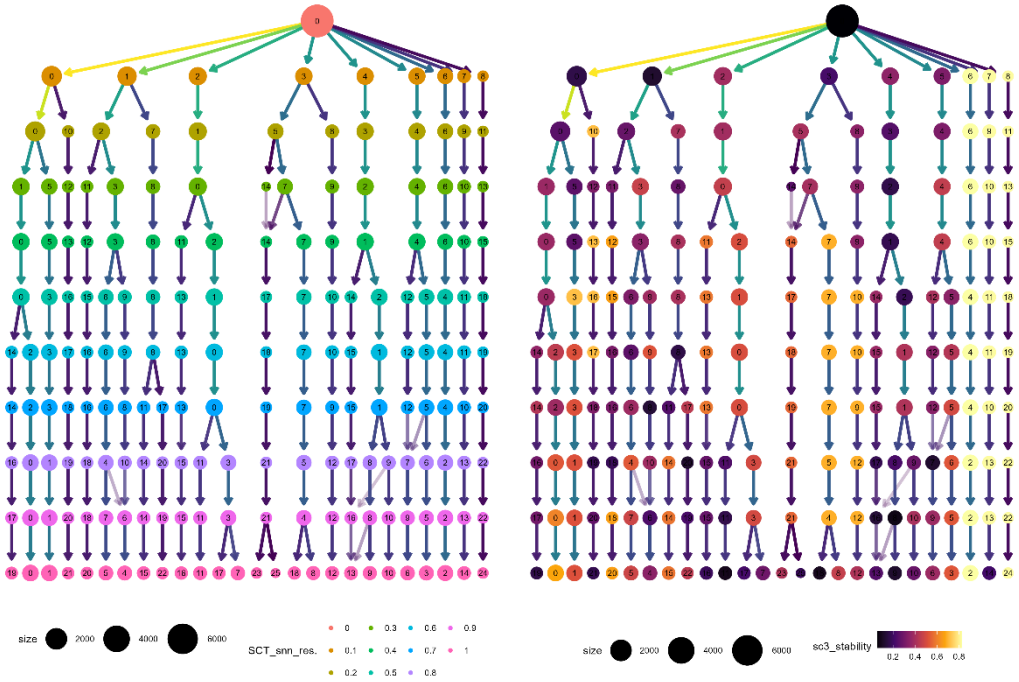


# Dimensionality reduction & Clustering

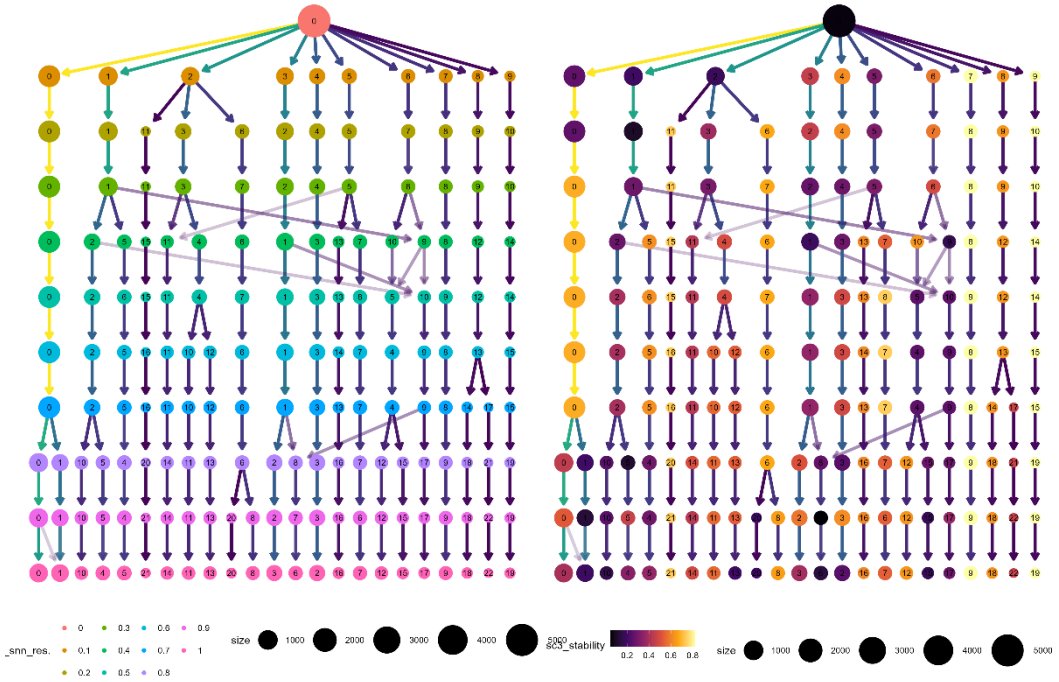
# Principal Component Analysis
# Healthy VS malignant



# Clustering



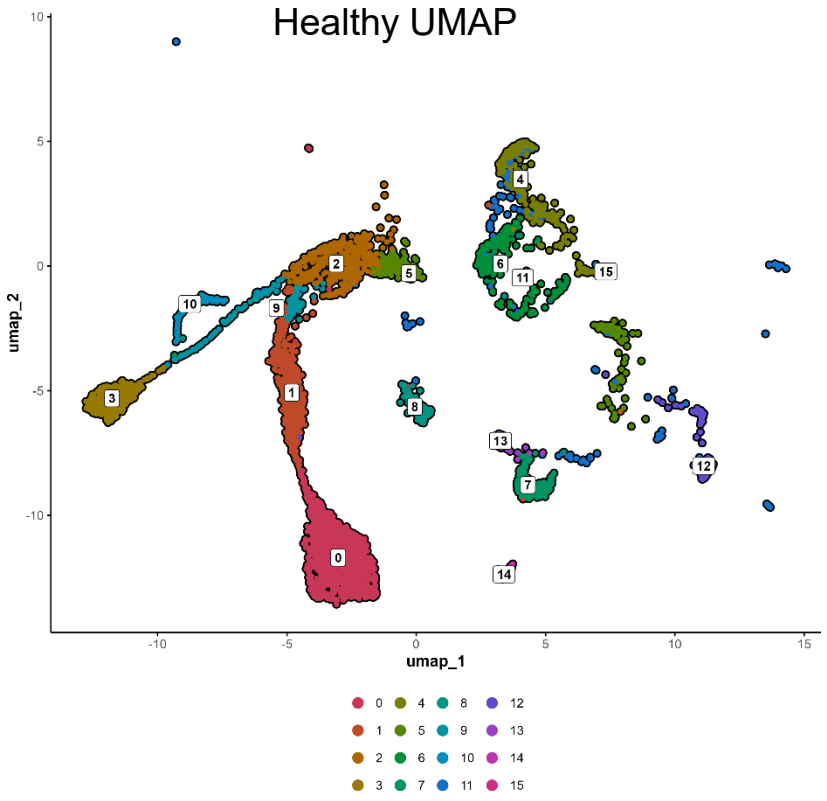

Malignant clustering tree

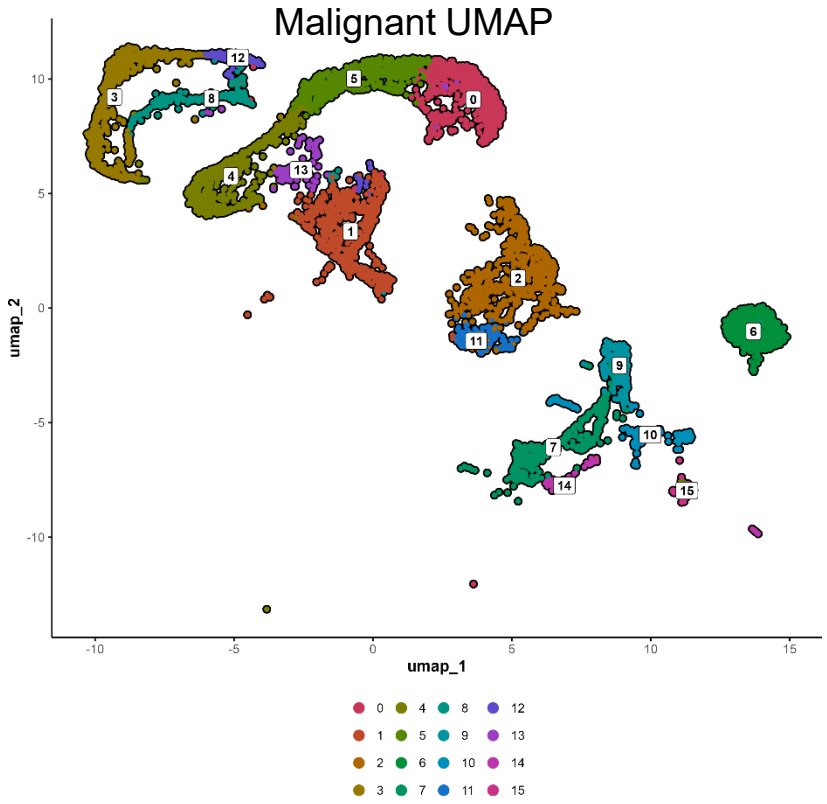



healthy clustering tree

- The data was split based on the condition into healthy subset and malignant subset
- Each subset was clustered and annotated individually
- Resolution 0.4 was selected for both the healthy and malignant

# Clusters visualization



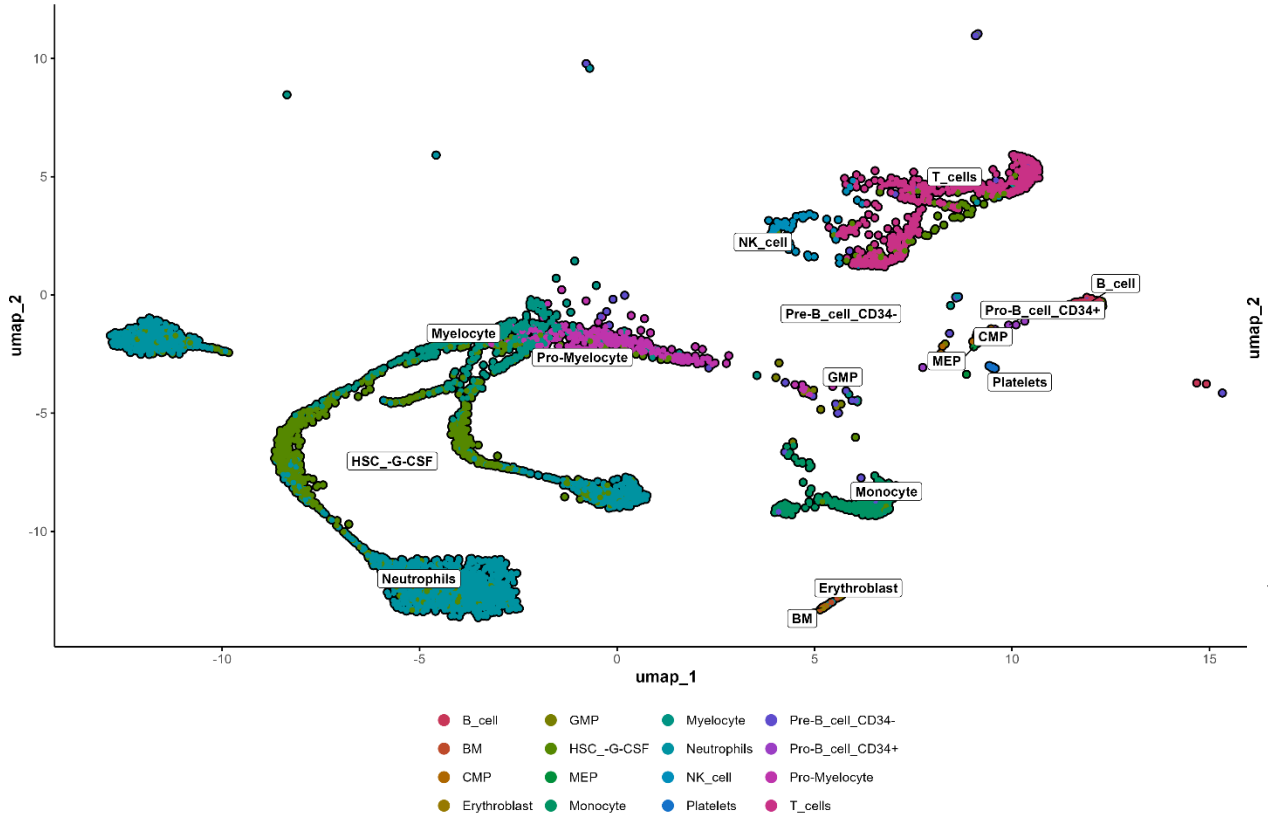




- Total of 16 cluster for each condition was visualized
- Some distant and lone cells can also be visualized at each condition which can be due to issues with the QC step

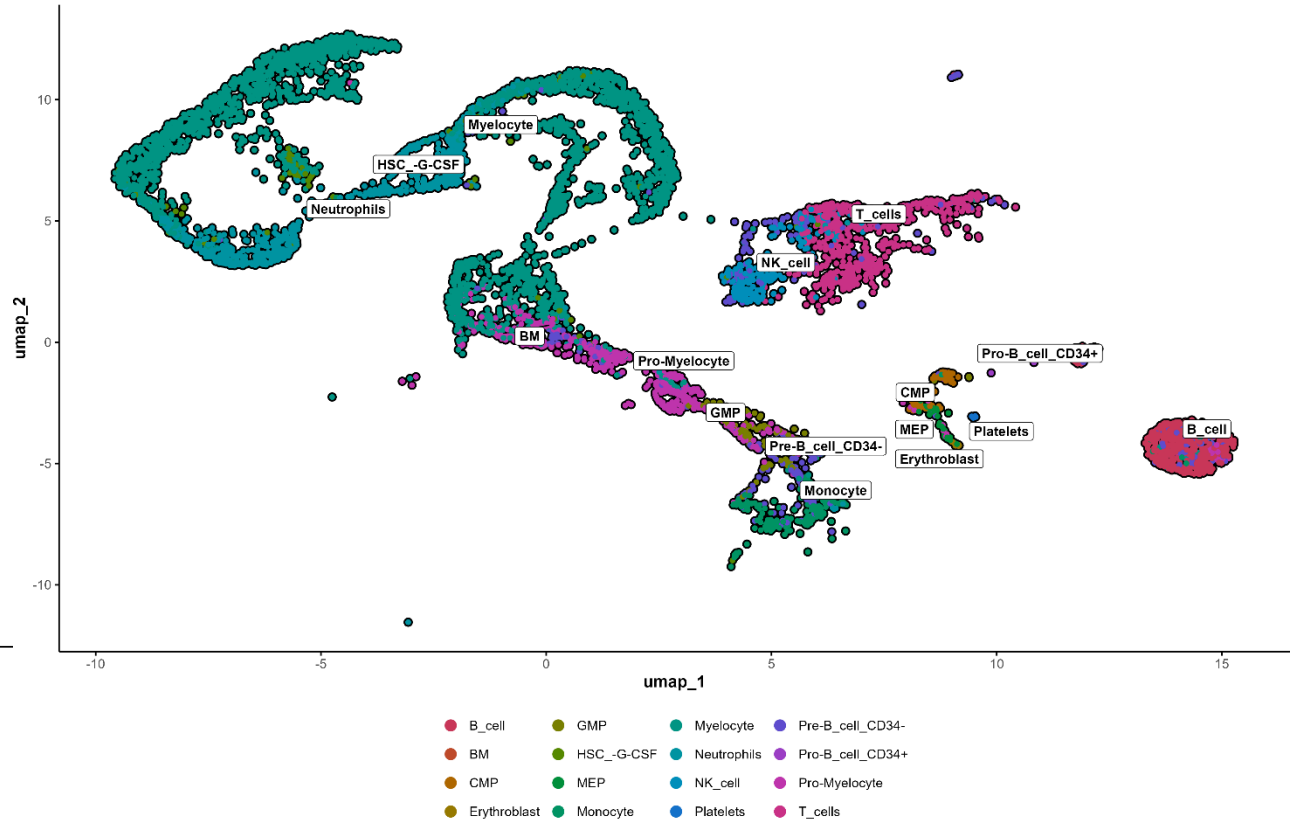
# Cell Type Annotation (SingleR)

# Cell Type Annotation (SingleR)

Healthy Cell Types



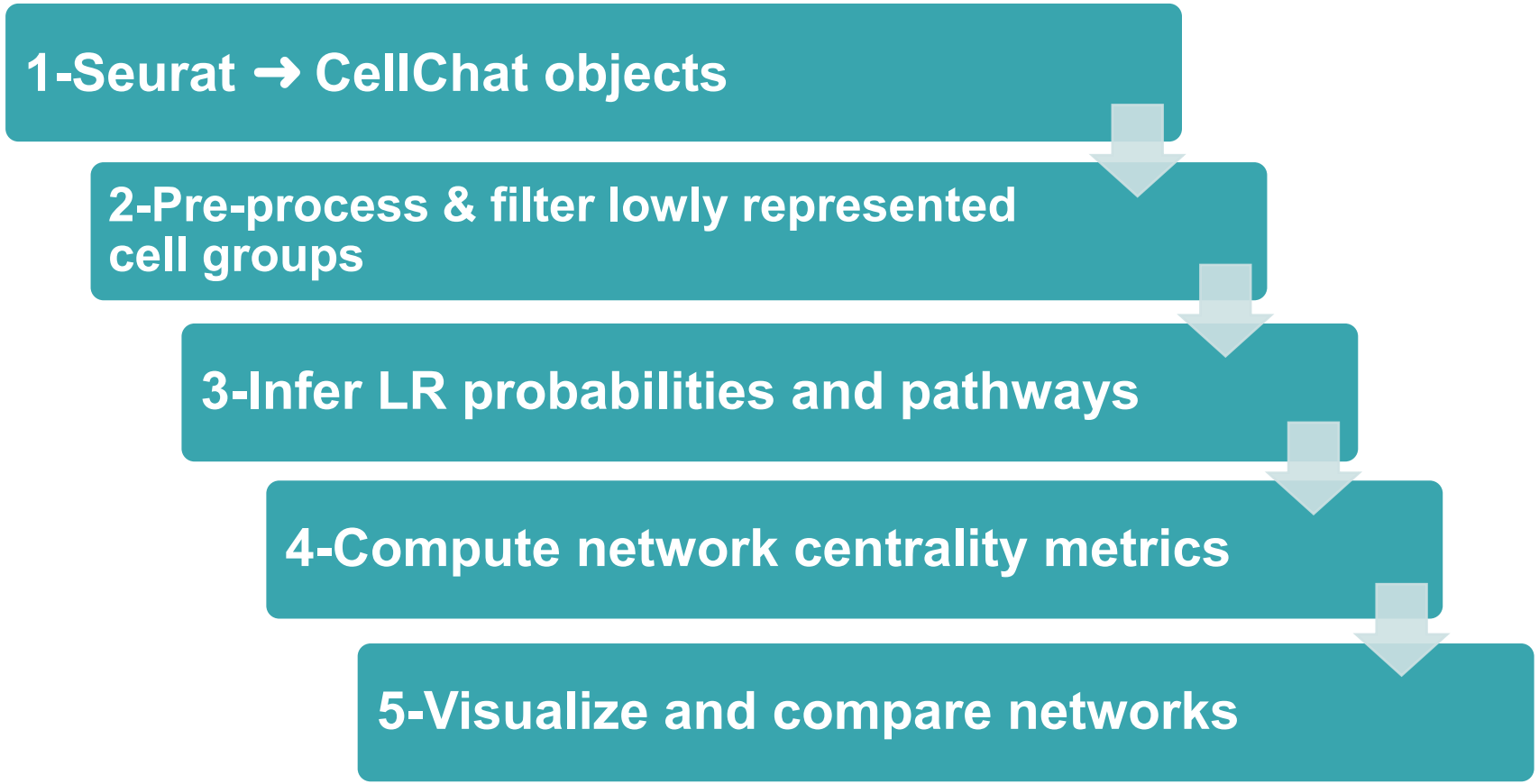

Malignant Cell Types




- Annotation was performed as cluster-based annotation and individual cell-based annotation
- For the cluster based, there were a lot of issues due to important cell types being covered in the same cluster by other cell types
- For the cell-based annotation, we were left with 16 different cell types that were the same between the 2 groups

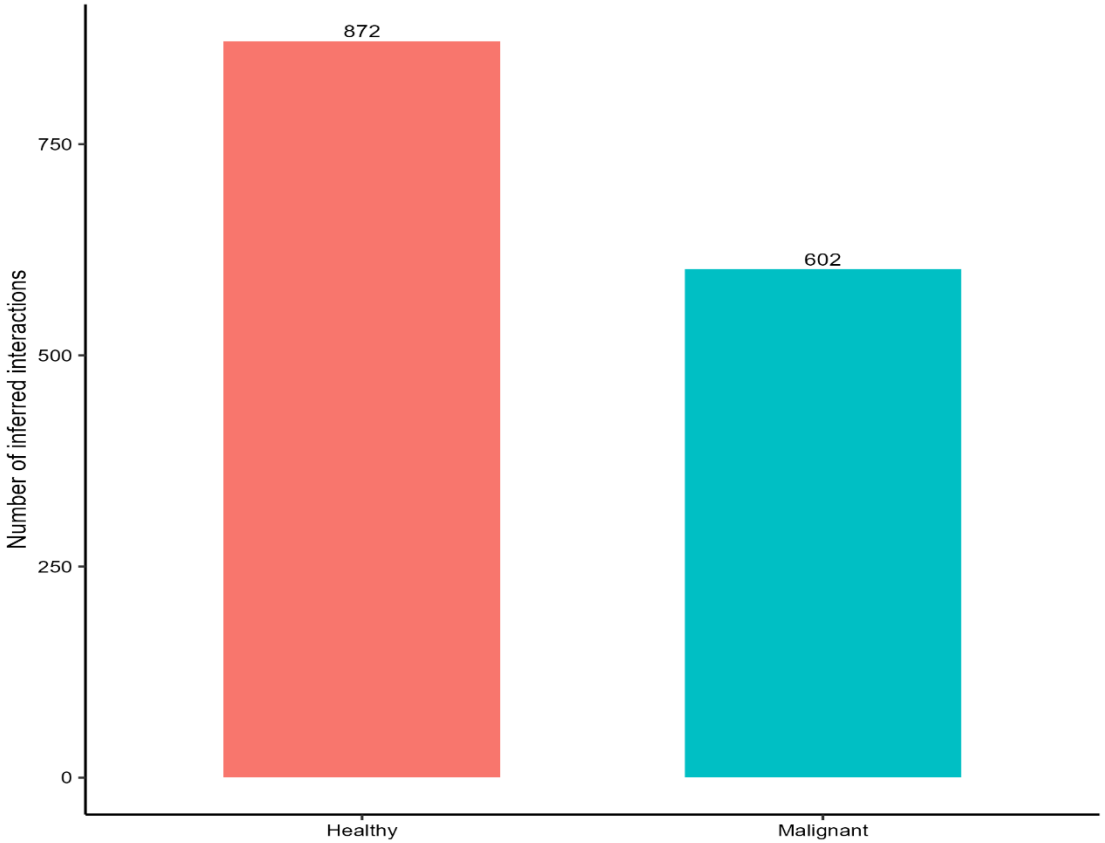
# Cell-Cell Communication Analysis (CellChat)

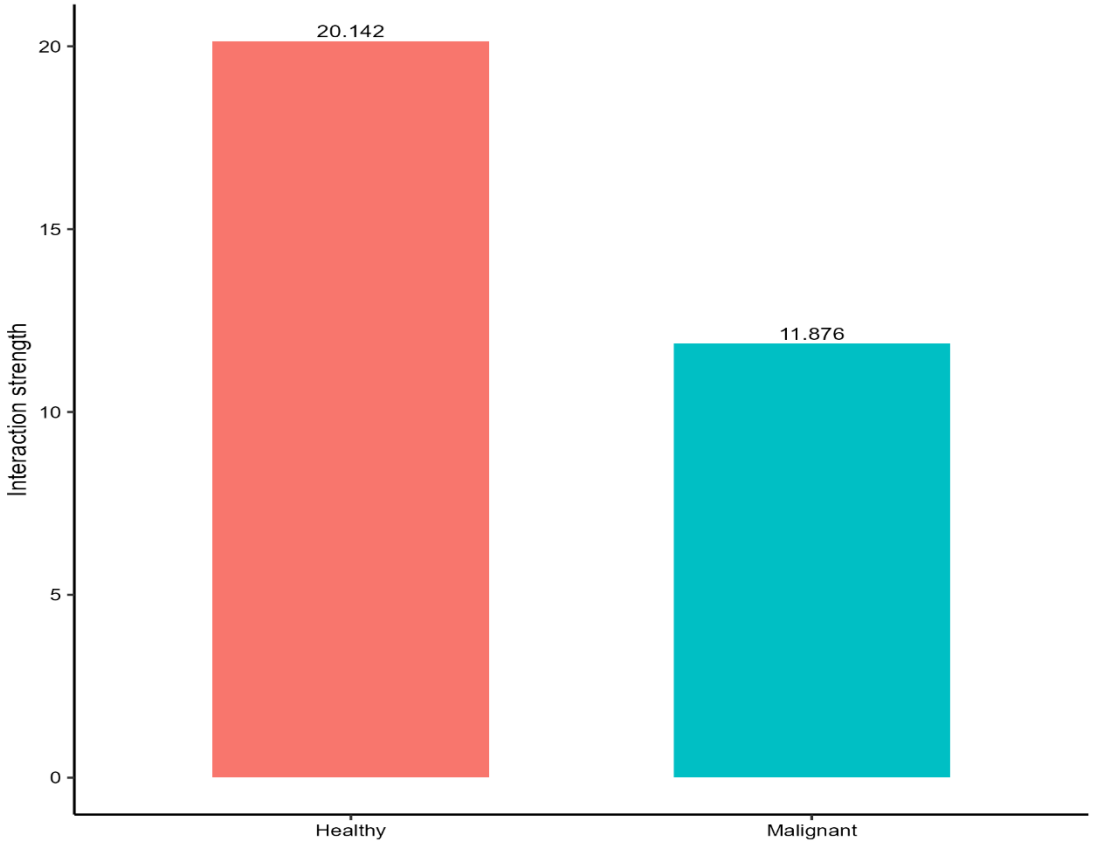
# Workflow Overview



# Total Interaction Counts & Weights

```
> chat_healthy
An object of class CellChat created from a single dataset
 18023 genes.
 5256 cells.
CellChat analysis of single cell RNA-seq data!
> chat_malignant
An object of class CellChat created from a single dataset
 18023 genes.
 7349 cells.
CellChat analysis of single cell RNA-seq data!
```



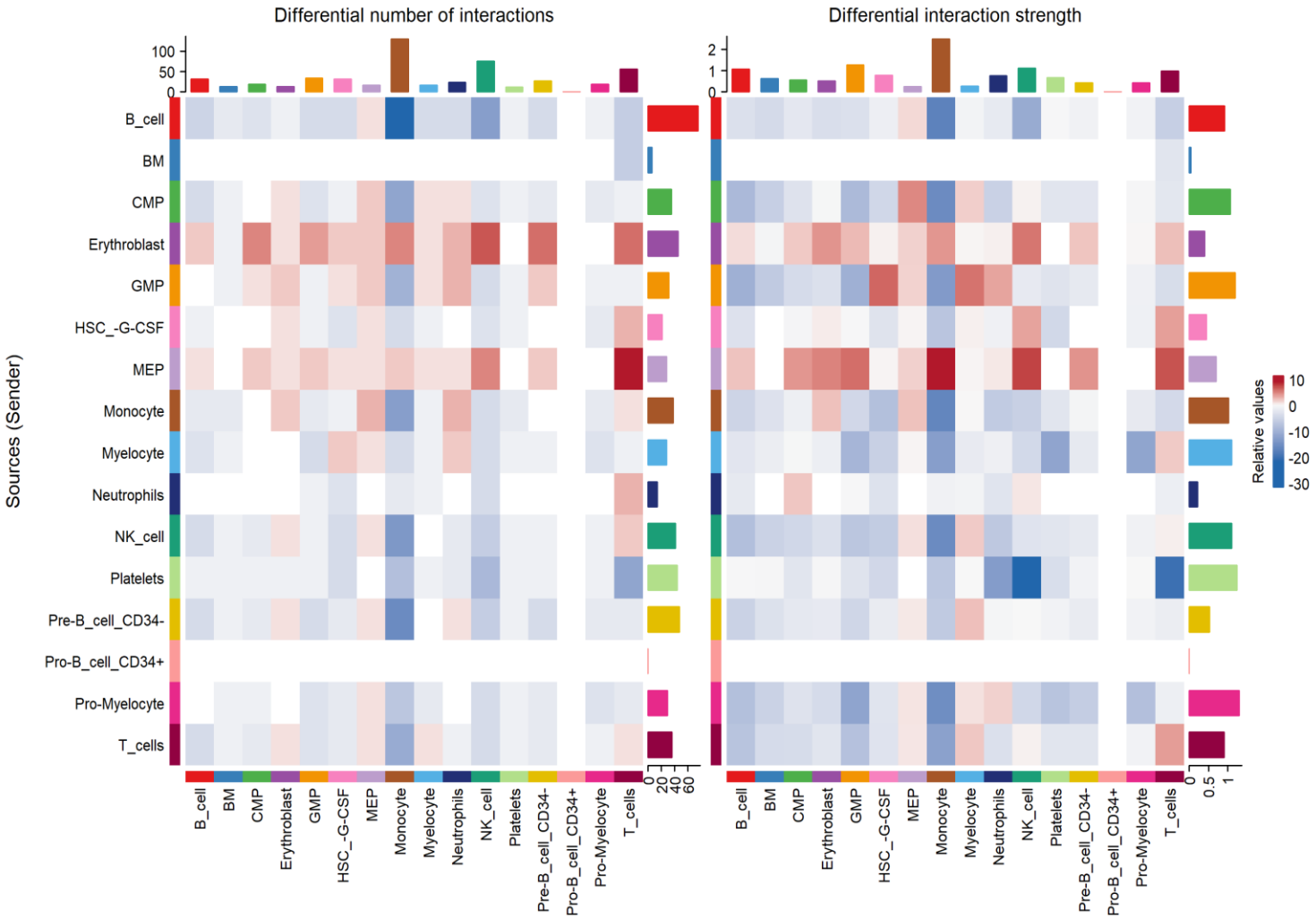

**31%**

**Reduction in interaction count in the malignant sample**

**41%**

**Reduction in interaction strength in the malignant sample**

# Interaction Counts and Weights Heatmap




Malignant > Healthy (Red):

- Erythroblasts, GMPs, and MEPs show increased outgoing communication in malignant marrow, most prominently toward CMPs and Monocytes; GMPs and MEPs also exhibit stronger autocrine signaling.

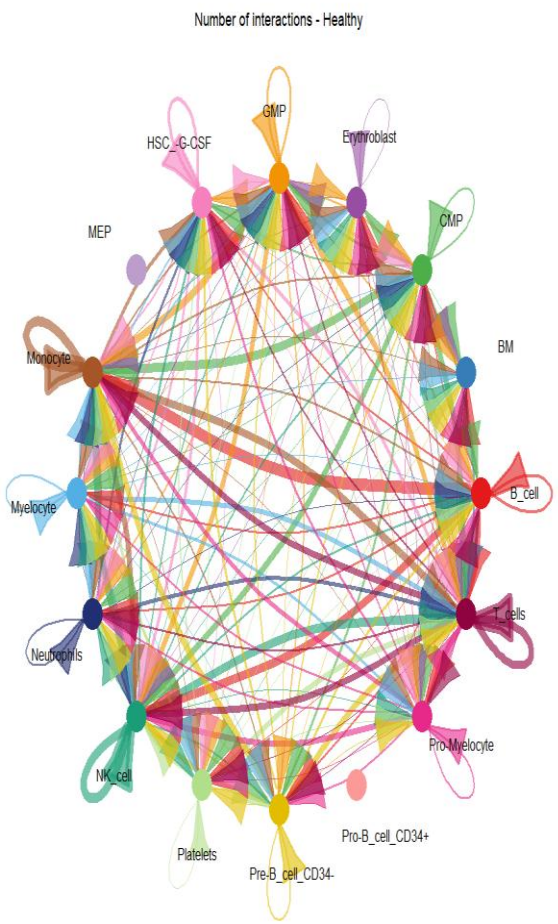
Healthy > Malignant (Blue):

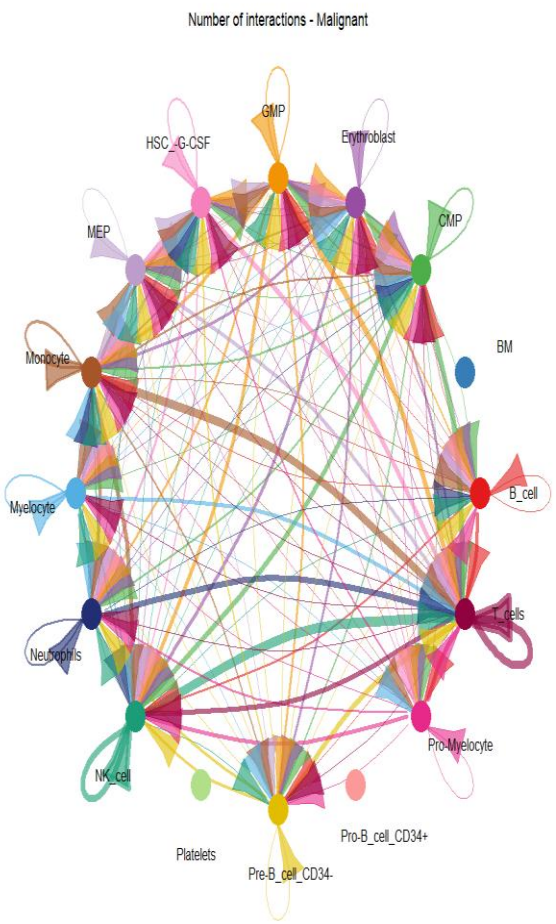
- Some cell types, particularly Neutrophils and Myelocytes, show reduced signaling in the malignant sample, indicating their communication activity is actually higher in healthy marrow.

Overall, malignancy suppresses the majority of intercellular signals, despite some cells showing increased signaling in the malignant sample.

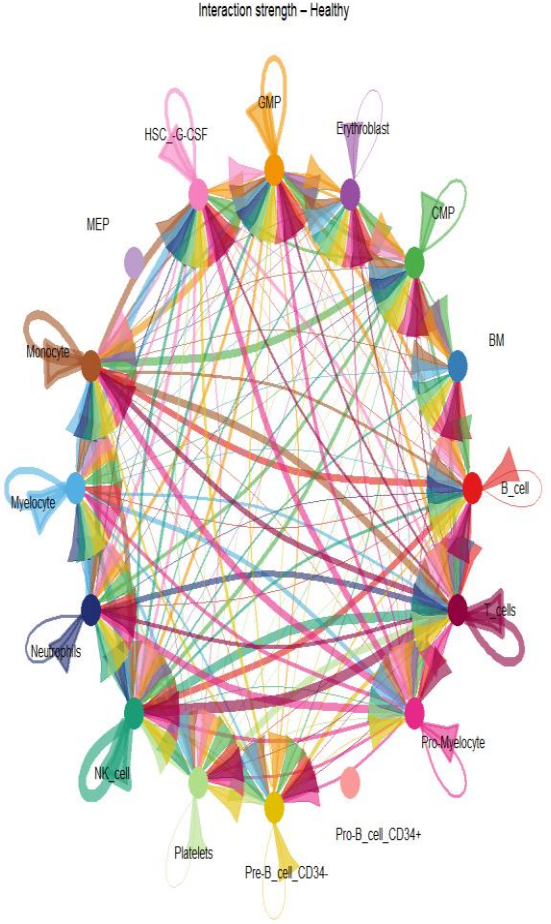
# Total Interaction Count & Strength (Healthy vs Malignant)

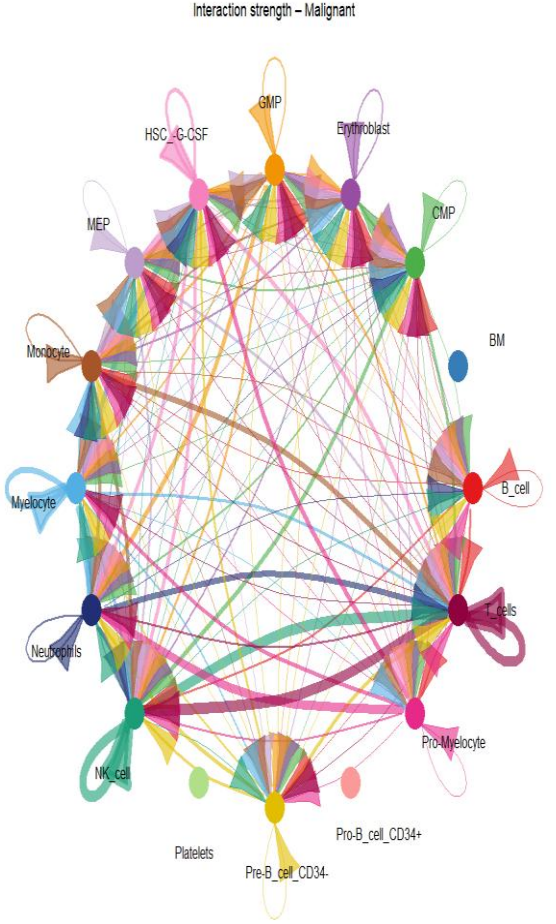
## Circle interactions count




## Circle interactions weights



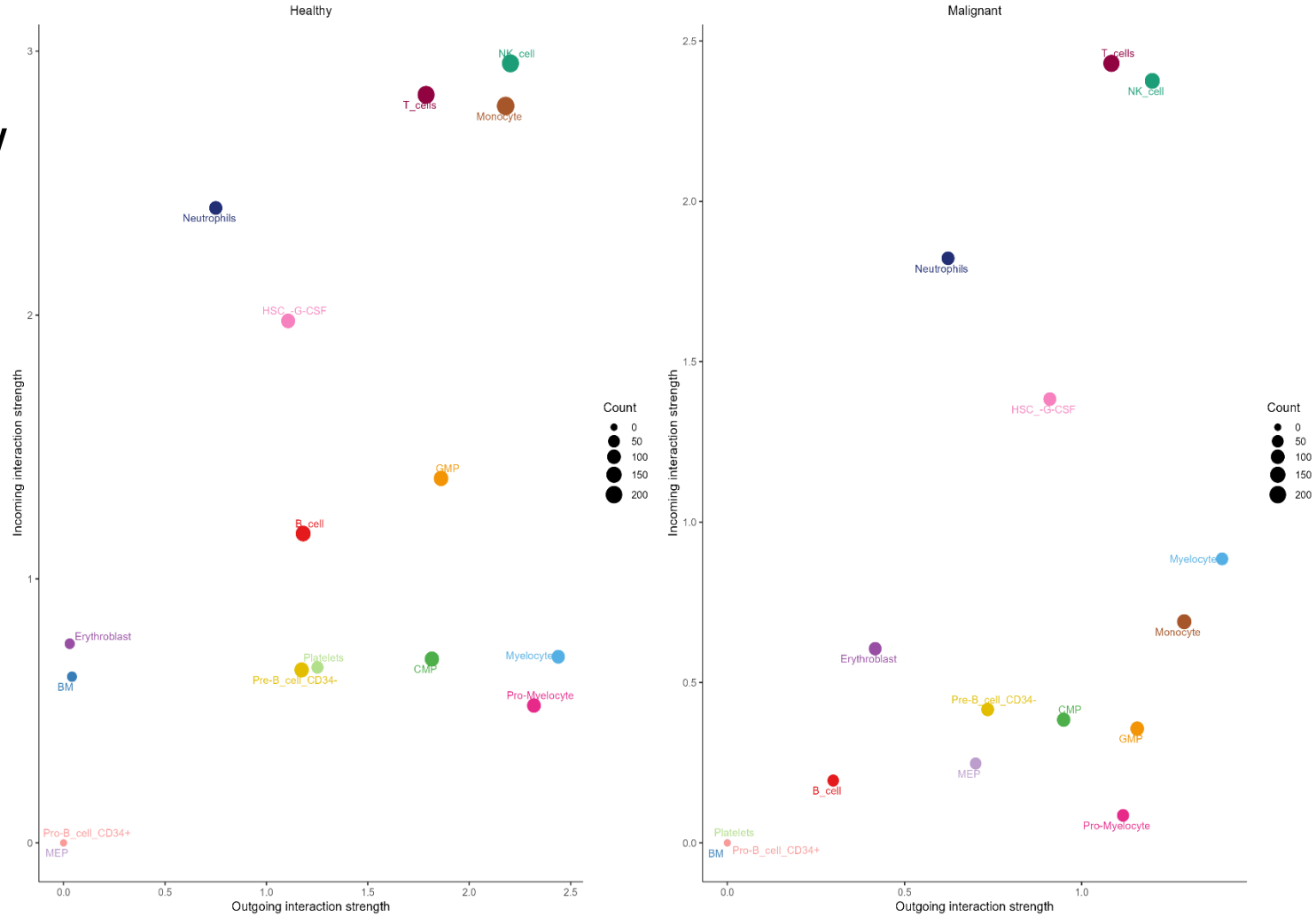
# Sender vs Receiver Roles

**signaling role scatter plot**

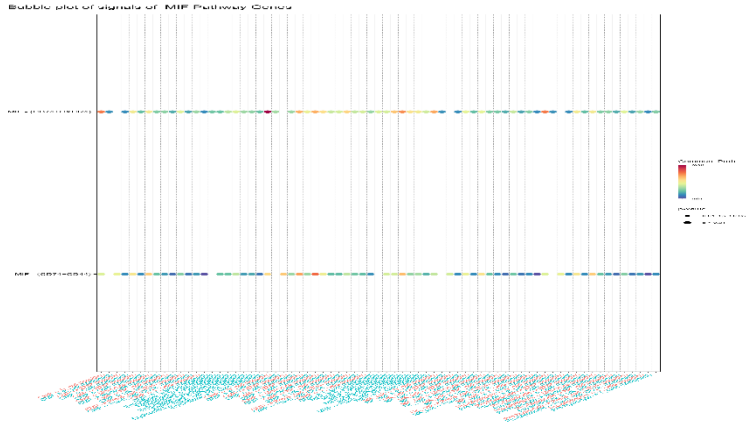
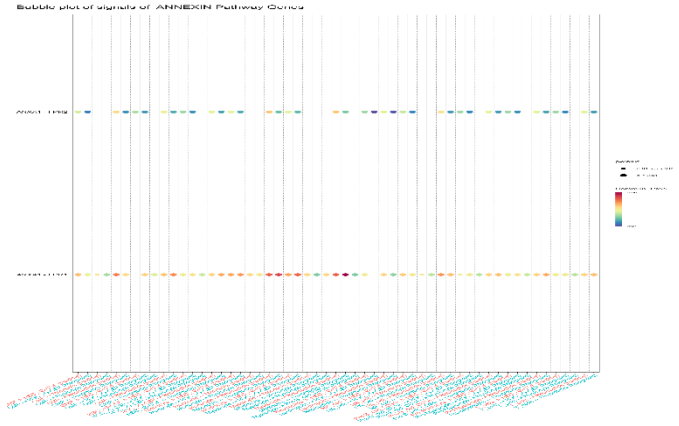
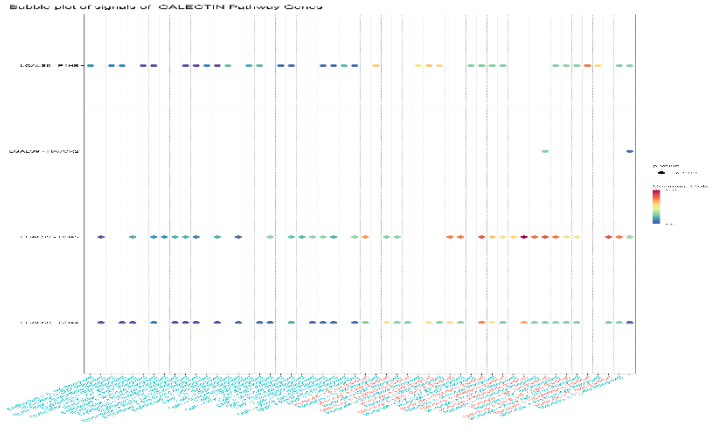
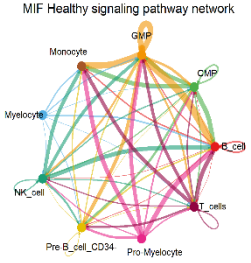
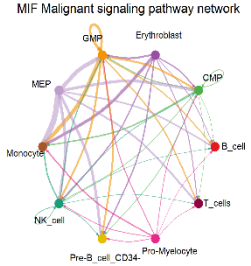
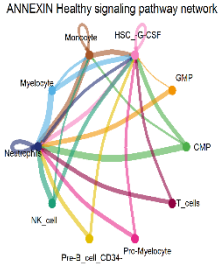
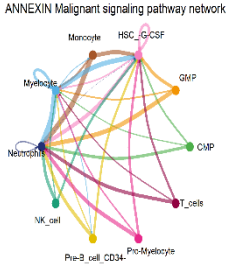
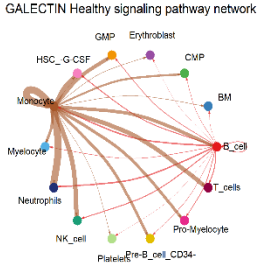
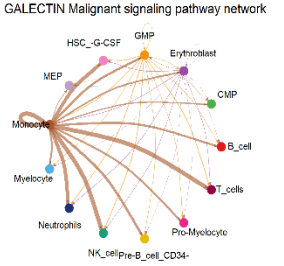
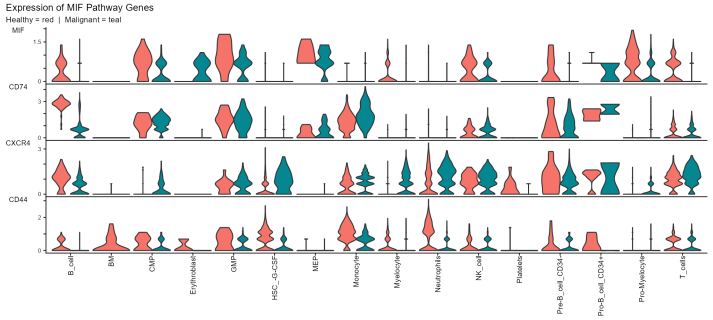
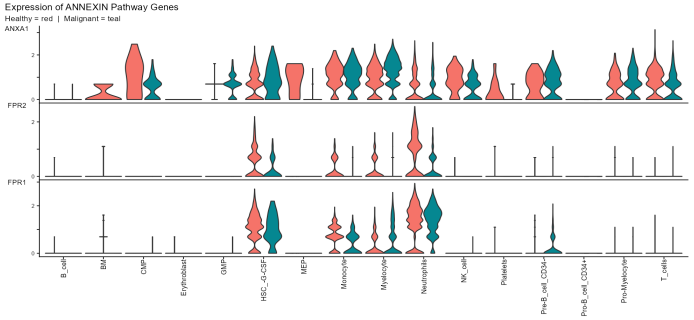
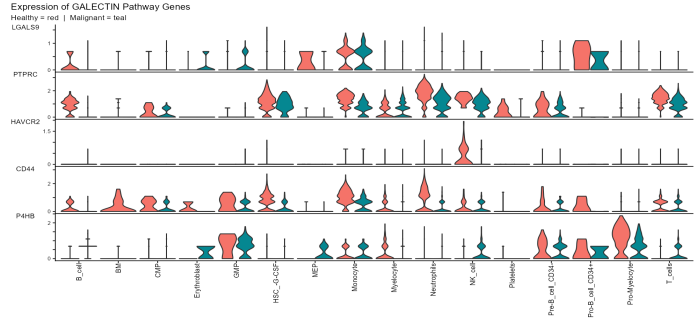
- **Left:** Healthy bone marrow
- **Right:** Malignant bone marrow

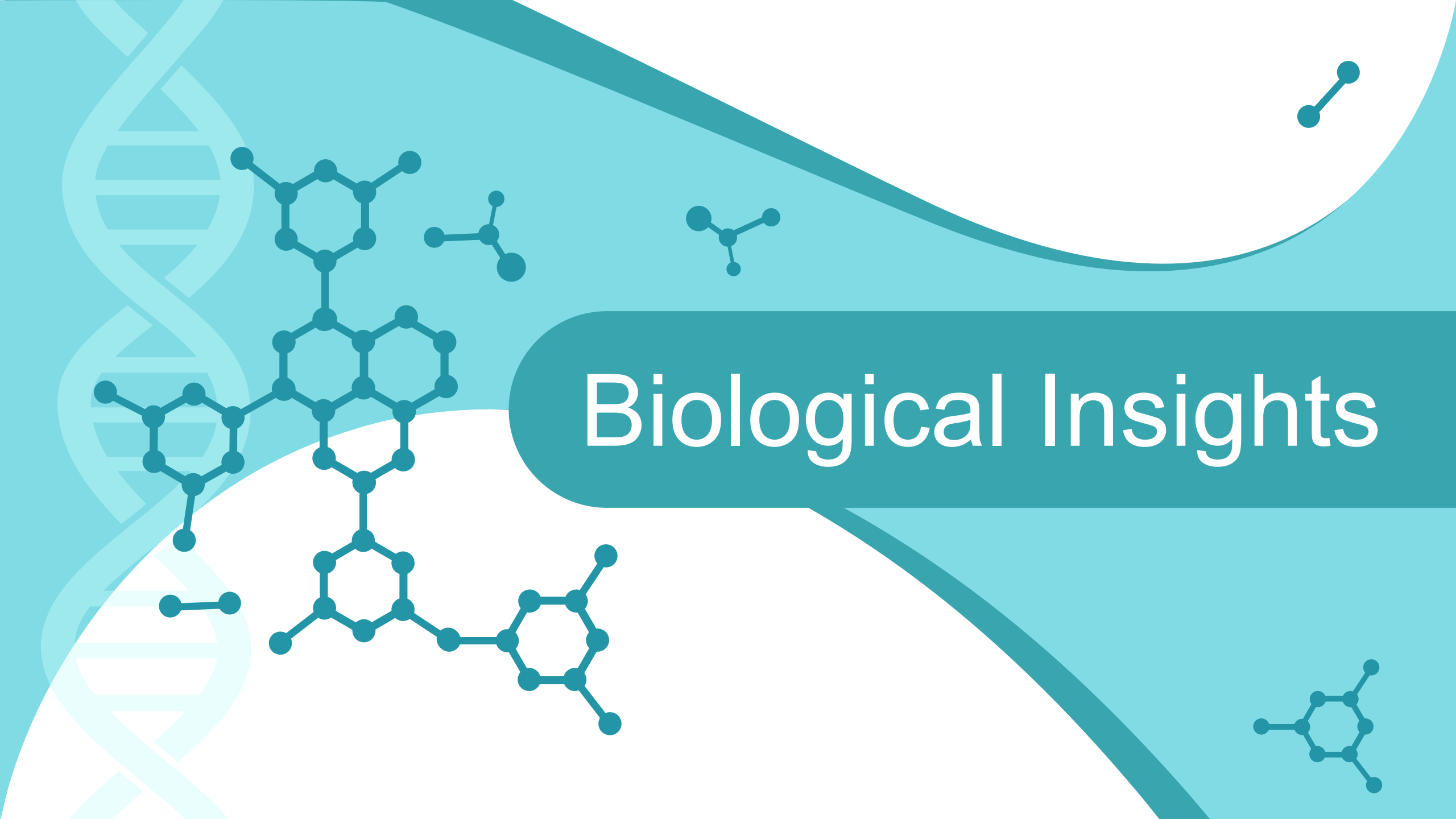
•Global collapse of incoming and outgoing strength in malignancy (all points shift down/left)

•T and NK cells remain the top communicators; Neutrophils still high for incoming signals, while Monocyte output is reduced.



# Selected Pathways: Healthy vs Malignant

| Plot Type | MIF | ANNEXIN | GALECTIN |
|---|---|---|---|
| Bubble Plot |  |  |  |
| Hierarchy |   |   |   |
| Violin |  |  |  |


Biological Insights

# Comparison of Selected Signaling Pathways

| Feature | MIF | ANNEXIN | GALECTIN |
|---|---|---|---|
| Primary Ligand → Receptor | MIF → CD74 / CXCR4, CD44 | ANXA1 → FPR1 / FPR2 | LGALS9 → CD44, CD45, P4HB |
| Main Senders | MEP, GMP, erythroblasts | Myelocytes, monocytes, GMP | Monocytes (dominant), GMP |
| Key Targets | Monocytes, GMP, NK, pre-B cells | Neutrophils, HSC-G-CSF, myelocytes | NK, T cells, monocytes, neutrophils |
| Change vs Healthy | Weaker (0.51x); share 13% to 11% | Weaker (0.81x); share 12% to 16% | Weaker (0.76x); more significant cell pairs |
| Functional Theme | Inflammation, M2 polarization (reported) | Anti-inflammatory, cell-survival signal | T-cell exhaustion / apoptosis |
| Immune-evasion Angle | Reported to skew macrophages; reduced here | Sustains suppressive myeloid niche | Disarms cytotoxic T-cell response |

# Collective Impact on Tumor Micro-Environment

- **Bubble & circle plots:** all three pathways are weaker in malignancy (0.5-0.8x).
- **Hierarchy plots:** ANNEXIN & GALECTIN keep a larger share of a smaller network.
- **Violin plots:** most pathway genes lower in malignancy.

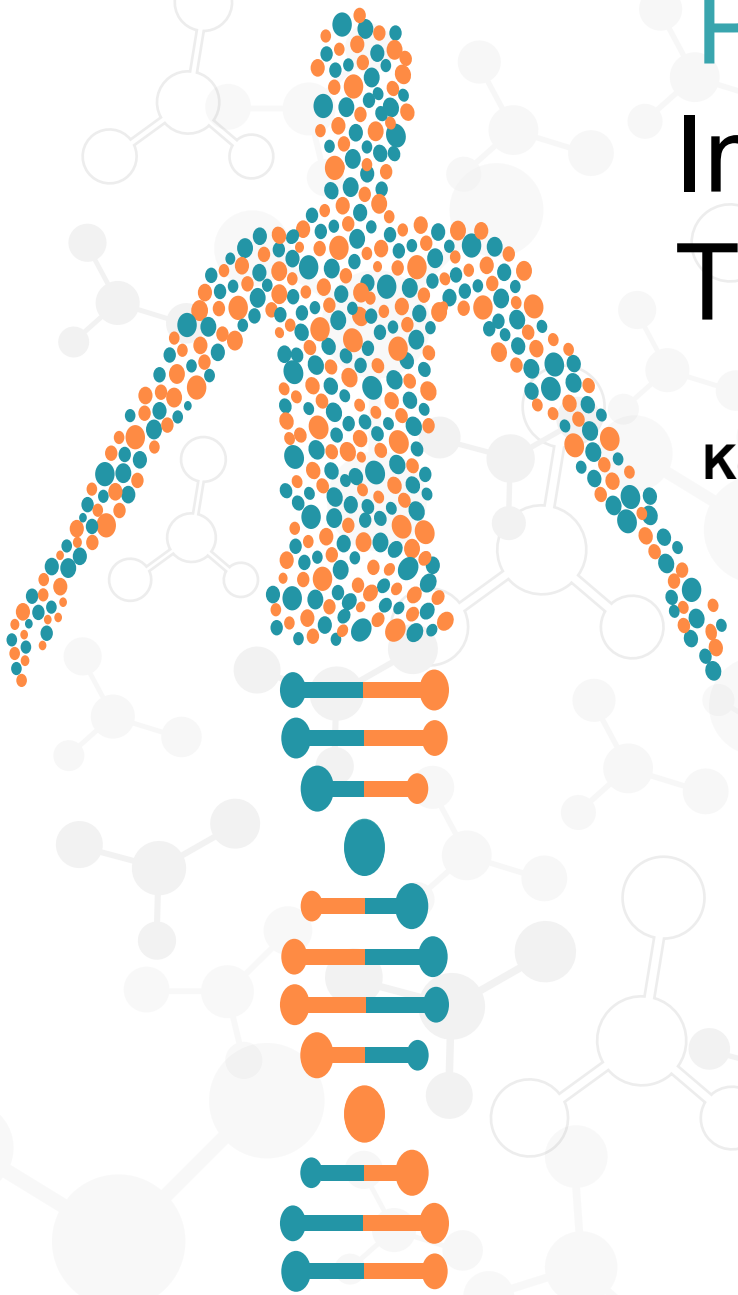
**01 Synergistic outcome**

- **MIF** is reported to prime M2-like macrophages, but was reduced here.
- **ANNEXIN** dampens pro-inflammatory cues → protects myeloid suppressors.
- **GALECTIN** drives T-cell dysfunction → weakens adaptive immunity.



**Network evidence (CellChat) 02**

❖ Communication is globally suppressed in malignant marrow; ANNEXIN & GALECTIN are relatively preserved, making them **candidate immune-regulatory axes** for further study.



# Proposed Mechanisms of Immune Evasion & Tumor Support

## Key Pathway-Driven Mechanisms

### 1. T-Cell Exhaustion & Suppression

LGALS9 from monocytes binds CD44/CD45 on T and NK cells, which can lead to dysfunction & weakened anti-tumor immunity.

### 2. M2-Like Macrophage Polarization

MIF-CD74 is reported to promote M2 polarization; MIF signaling was reduced in this sample.

### 3. Inhibitory Microenvironment

ANNEXIN keeps a larger share of signaling and may dampen pro-inflammatory cues and limit immune activation.

### 4. Stromal-Driven ECM Remodeling

Stromal cells such as fibroblasts were not captured in this dataset, so ECM-driven niche protection could not be assessed.



# Thank You